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A systematic evaluation of filter Unsupervised Feature Selection methods

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ABSTRACT

Unsupervised Feature Selection (UFS) has aroused great interest in the last years because of its practical significance and application on a large variety of problems in expert and intelligent systems where unlabeled data appear. Specifically, Unsupervised Feature Selection methods based on the filter approach have received more attention due to their efficiency, scalability, and simplicity. However, in the literature, there are no comprehensive studies for assessing such UFS methods when they are applied, under the same conditions, to a wide variety of real-world data. To fill this gap, in this paper, we present a comprehensive empirical and systematic evaluation of the most popular and recent filter UFS methods, evaluating their performance in terms of clustering, classification, and runtime. The filter methods used in our study were applied on 50 datasets from the UCI Machine Learning Repository and 25 high dimensional datasets from the ASU Feature Selection Repository. To evaluate if the outcomes obtained by the assessed methods are statistically significant, the Friedman test and Holm post hoc procedure were applied in the clustering and classification results. From our experiments, we provide some practical guidelines and insights for the use of the filter UFS methods analyzed in our study.

1. Introduction

Feature Selection (Liu & Motoda, 1998; Guyon, Elisseeff, & De, 2003; Liu & Motoda, 2007), also known as attribute selection, is the process of identifying and selecting those relevant and, in some cases, non-redundant features that allow maintaining or even improving the performance of learning models without the need of using the whole set of original features. Feature Selection is useful in computational learning tasks such as classification, regression, or clustering because besides reducing the storage and processing requirements, it helps to avoid the curse of dimensionality (Zhao et al., 2011) and allows to generate compact models with better generalization capability (Pal & Mitra, 2004) in expert and intelligent systems applications. All these characteristics make feature selection an interesting research area, wherein the last decades, numerous feature selection methods have been introduced.

According to the availability of information in the datasets, Feature Selection methods can be classified as supervised (Kotsiantis, 2011; Tang, Alelyani, & Liu, 2014), semi-supervised (Sheikhpour, Sarram, Gharaghani, & Chahooki, 2017) and unsupervised (Fowlkes, Gnanadesikan, & Kettenring, 1988; Dy & Brodley, 2004; Alelyani, Tang, & Liu, 2013). The former require a set of labeled data (supervised dataset) in order to perform the feature selection; this label, assigned to each object in the dataset, can be a category, an ordered value, or a real value (depending on the specific task). Semi-supervised methods only require that some objects be labeled. On the other hand, Unsupervised Feature Selection (UFS) methods do not require a supervised dataset and they are very useful nowadays because much of the information that is generated in many applications such as text mining, image retrieval, bioinformatics, and social media, to name a few, is of unsupervised nature. Moreover, according to Guyon et al. (2003), Niijima and Okuno (2009) and Devakumari and Thangavel (2010), UFS methods have two important advantages. (1) they are unbiased and perform well when prior knowledge is not available, and (2) they can reduce the risk of data over-fitting in contrast to supervised feature selection methods that may be unable to deal with a new class of data.

According to the evaluation criterion used for selecting features, UFS methods can be divided into three main approaches: filter, wrapper, and hybrids (Alelyani et al., 2013; Dong & Liu, 2018). Filter methods evaluate features based on intrinsic properties of the data, without using any clustering algorithm that could guide the search of relevant features. The main characteristic of filter methods is their speed and scalability. On the other hand, wrapper methods (Kim, Street, & Menczer, 2002; Dy & Brodley, 2004;

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Law, Figueiredo, & Jain, 2004; Breaban & Luchian, 2011; Dutta, Dutta, & Sil, 2014; Marbac & Sedki, 2017) evaluate feature subsets using the results of a specific clustering algorithm. Methods developed under this approach are characterized by finding features subsets that contribute to improving the quality of the results of the clustering algorithm used for the selection. However, they usually have a high computational cost, and they are limited to be used in conjunction with a particular clustering algorithm. Finally, hybrid methods (Dash & Liu, 2000; Hruschka, Hruschka, Covões, & Ebecken, 2005; Li, Lu, & Wu, 2006; Solorio-Fernández, Carrasco-Ochoa, & Martínez-Trinidad, 2016) try to exploit the qualities of both approaches, filter, and wrapper, trying to have a good compromise between efficiency (computational effort) and effectiveness (quality in the associated objective task when using the selected features). Nevertheless, in hybrid methods, typically, the filter and wrapper approaches are not truly integrated with each other, which may lead to lower classification performance.

Due to their advantages in terms of speed and scalability of filter methods, in the literature, a large number of methods based on different strategies have been developed under this approach (Solorio-Fernández, Carrasco-Ochoa, & Martínez-Trinidad, 2020). However, there are no empirical comprehensive studies[1] of such methods that allow the user to determine which particular UFS method could be more suitable or appropriate for a given problem. To fill this gap, in this paper, following the taxonomy presented in Solorio-Fernández et al. (2020), we have selected and compared 18 of the most relevant (those with more citations) and recent filter UFS methods (both univariate and multivariate) of the state-of-the-art over several datasets of different nature (numerical, non-numerical and mixed), size, and dimensionality (small, medium and high), taken from public repositories. The contribution of this work is twofold. On the one hand, the goal is to carry out an empirical and systematic evaluation of the performance of these methods in classification and clustering tasks under the same conditions. On the other hand, we aim to perform an analysis based on the experimental results that allows getting some guidelines for choosing a priori which method would be the best option. To the best of our knowledge, this is the first paper in the literature that performs a comprehensive comparison of the most relevant and recent (univariate and multivariate) unsupervised filter feature selection methods.

The rest of this paper is organized as follows, in Section 2, the 18 filter Unsupervised Feature Selection methods analyzed in our study will be described. The experimental methodology of our study will be described in Section 3. The systematic evaluation, as well as the analysis of the results and discussion, are given in Section 4. Finally, in Section 5, we present our conclusions and some future work directions.

2. Filter UFS methods

According to Alelyani et al. (2013), UFS methods based on the filter approach can be categorized as *univariate* and *multivariate*. The former, also known as ranking based UFS methods use some criteria to evaluate each feature in order to get an ordered list (ranking) of features, where the final feature subset is selected according to this order. Such methods can effectively identify and remove irrelevant features, but they are unable to remove redundant ones since they do not take into account possible dependencies among features. On the other hand, multivariate filter methods evaluate the relevance of the features jointly rather than individually. Multivariate methods can handle redundant and irrelevant features; thus, in many cases, the accuracy reached by learning algorithms using the subset of features selected by multivariate methods is better than the one achieved by using univariate methods (Tabakhi, Najafi, Ranjbar, & Moradi, 2015).

In the following, we briefly describe the filter (univariate and multivariate) UFS methods evaluated in this paper.

2.1. Univariate filter methods

Within univariate filter methods, there are different criteria commonly used for assessing the relevance of each feature. Among the most employed are those based on Statistics, Information Theory, and Spectral Analysis (manifold learning). Next, we will describe some of the most relevant methods in these categories.

Variance: Variance (Dy & Brodley, 2004; Liu, Kang, Yu, & Wang, 2005; Theodoridis & Koutroumbas, 2008) is the most simple univariate filter method for evaluating the importance of a single feature into an unsupervised context. The idea is that features with higher variance are more relevant for building good cluster structures. This method ranks features by their variance in the dataset.

SUD: Sequential backward selection method for Unsupervised Data (Dash, Liu, & Yao, 1997) weighs features using a similarity entropy measure based on distance. This entropy measure is defined as the total entropy induced from a similarity matrix W, where the elements of this matrix contain the similarity between pairs of objects in the dataset. The idea is to calculate the entropy that is generated after removing one feature at a time from the whole set of features. In this way, a feature ranking ordered from the most relevant (which gives a high entropy after removing it) to the least relevant feature (lowest entropy) is obtained.

LS: Laplacian Score (He, Cai, & Niyogi, 2005) is a spectral feature selection method that measures the relevancy of a feature based on the Laplacian matrix derived from the similarities among objects. The idea is to weight each feature measuring how this feature reflects the underlying manifold structure (He & Niyogi, 2004); i.e., those features most preserving the pre-defined graph structure represented by the Laplacian matrix are the most relevant. This method gets a feature ranking according to the Laplacian score associated with each feature.

SVD-Entropy: SVD-Entropy (ranking-based version) (Varshavsky, Gottlieb, Linial, & Horn, 2006) selects those features that best represent the data, measuring the entropy of the original data matrix through its singular values (Alter & Alter, 2000).The contribution of each feature to the entropy (CE) is evaluated through a leave-one-out comparison, and the features are sorted according to their respective CE values.

SPEC: SPECtrum decomposition (Zhao & Liu, 2007) is a spectral feature selection method that evaluates the relevance of a feature by its consistency[2] with the structure of the graph induced from the similarities among objects. Feature evaluation is performed by measuring the consistency between each feature and those nontrivial eigenvectors of the Laplacian matrix (Chung, 1997) through different score functions. The final result is an ordered list sorted from the most to the least consistent (relevant) feature.

USFSM: Unsupervised Spectral Feature Selection Method for mixed data (Solorio-Fernández, Martínez-Trinidad, & Carrasco-Ochoa, 2017) is a ranking-based feature selection

[1] Except in Solorio-Fernández, Carrasco-Ochoa, and Martínez-Trinidad (2018), where we report a preliminary study with just a few univariate ranking based filter UFS methods on high dimensional datasets.

[2] A feature is consistent if it takes similar values for the objects are close to each other, and dissimilar values for the objects are far apart.

method based on Spectral Feature Selection (Zhao et al., 2011). The idea is to rank the features according to their consistency in the dataset by analyzing the changes in the spectrum distribution (spectral gaps) of the Normalized Laplacian matrix when each feature is excluded from the whole set of features separately. Features are sorted in descending order according to their respective spectral gaps values.

2.2. Multivariate filter methods

Among multivariate filter methods, we can distinguish three main groups: Statistical, Bio-inspired, and Spectral/Sparse-Learning-based methods. The former, as its name suggests, includes UFS methods that perform the selection using statistical measures such as variance–covariance and linear correlation. The second group, on the other hand, includes UFS methods that use stochastic search strategies based on the swarm intelligence paradigm for finding a good subset of features, which satisfies some criterion of quality. Finally, the third group includes those UFS methods based on Spectral Analysis or a combination of Spectral Analysis and Sparse Learning. It is noteworthy that some authors (Saeys, Inza, & Larrañaga, 2007; Chandrashekar & Sahin, 2014; Jović, Brkić, & Bogunović, 2015; Li et al., 2016; Ang, Mirzal, Haron, & Hamed, 2016; Li et al., 2016) often call these last methods as *embedded* because feature selection is achieved as part of the learning process, commonly through the optimization of a constrained regression model. However, in this study, we will handle them as filter multivariate, since in addition to jointly evaluate features, the primary objective is to perform feature selection (or ranking) rather than finding the cluster labels, and the final output of these methods commonly does not depend on any clustering algorithm (Qian & Zhai, 2013). Moreover, we think that embedded methods could be considered as a sub-category inside the main approaches (i.e., filter, wrapper, and hybrid), not hindering the possibility of having embedded methods in the three approaches.

Some of the most relevant methods in the categories mentioned above are described below.

FSFS: Feature Selection using Feature Similarity (Mitra, Murthy, & Pal, 2002) introduces a statistical measure of dependency/similarity based on the variance–covariance to reduce feature redundancy. The idea of this method is partitioning the original set of features into clusters (using the dependency/similarity measure), such that those features in the same cluster are highly similar, while those in different clusters are dissimilar. Feature selection is accomplished iteratively based on the kNN principle, wherein each iteration (one for each feature), FSFS selects only one feature from each cluster to form the final feature subset.

MC: Mutual-Correlation (Haindl, Somol, Ververidis, & Kotropoulos, 2006) is a fast and straightforward method that can be used either in a supervised or unsupervised mode. The idea is to evaluate all mutual correlations for all feature pairs; then, the feature with the largest average mutual correlation with all other features is removed, and the process is repeated for the remaining features until a number of features, previously specified by the user, is reached.

MCFS: Multi-Cluster Feature Selection (Cai, Zhang, & He, 2010) captures the local manifold structure via spectral analysis (Luxburg, 2007) and then selects the features which can best preserve the clustering structure. Through the first k eigenvectors of the Laplacian matrix, MCFS measures the importance of the features by a regression model with a l_1-norm regularization (Donoho & Tsaig, 2008). For selecting the final feature subset, MCFS selects d features based on the highest absolute values of the coefficients obtained through the regression problem.

MRSF: Minimize the feature Redundancy for Spectral Feature selection (Zheng, Lei, & Huan, 2010) evaluates the features all together in order to eliminate redundant features. The idea is to formulate the feature selection problem as a multi-output regression problem (Friedman, Hastie, & Tibshirani, 2001), and the selection is performed by enforcing the sparsity applying the $l_{2,1}$-norm (Argyriou, Evgeniou, & Pontil, 2008). Moreover, in this work, an efficient algorithm based on Nesterov's method (Liu, Ji, & Ye, 2009) for solving the regression problem was also proposed. The final feature subset is selected based on the values of a weighted W matrix.

UDFS: Unsupervised Discriminative Feature Selection method (Yang, Shen, Ma, Huang, & Zhou, 2011) weighs features by simultaneously exploiting the discriminative information contained in the scatter matrices and feature correlations. UDFS addresses the feature selection problem by taking into account the trace criterion (Fukunaga, 1990) into a constrained regression problem optimized through an efficient optimization algorithm. Each feature f_i is ranked according to the corresponding w_i value in descending order, and the top-ranked features are selected.

NDFS: Nonnegative Discriminative Feature Selection (Li, Yang, Liu, Zhou, & Lu, 2012) performs feature selection exploiting the discriminative information and feature correlations in a unified framework. First, NDFS uses spectral analysis (Ng, Jordan, & Weiss, 2002) to learn pseudo-class labels (defined as non-negative real values). Then, a regression model with $l_{2,1}$ norm regularization (Argyriou et al., 2008) is formulated and optimized through a special solver also proposed in this work. The top p ranked features that are the most related to the pseudo-class labels are selected.

RRFS: Relevance Redundancy Feature Selection (Ferreira & Figueiredo, 2012) is a supervised/unsupervised filter method that selects features in two steps. First, the features are sorted according to a relevance measure (variance for the unsupervised version and the Fisher's Ratio or mutual information for the supervised one). Then, in the second step, following the order generated in the previous step, the features are evaluated using a feature similarity measure to quantify the redundancy between them. Afterward, the first p features with the lowest redundancy are selected.

JELSR: Joint Embedding Learning and Sparse Regression (Hou, Nie, Yi, & Wu, 2011; Hou, Nie, Li, Yi, & Wu, 2014) JELSR works with the same objective function as MRSF (Zheng et al., 2010), and it only differs in the construction of the Laplacian graph, since in this work, locally linear approximation weight (Roweis & Saul, 2000) is used to measure local similarities for building the Laplacian graph. The same authors introduced a later generalization of JELSR in Hou et al. (2014), where instead of using the Laplacian graph to characterize the structure of high dimensional data and then apply regression, a unify embedding learning and sparse regression framework was proposed.

UFSACO: Unsupervised Feature Selection based on Ant Colony Optimization (ACO) (Tabakhi, Moradi, & Akhlaghian, 2014) is a bio-inspired Unsupervised Feature Selection method based on the swarm intelligence paradigm (Beni & Wang, 1993; Dorigo & Gambardella, 1997). The main objective is to select feature subsets with low similarity among features (low redundancy) modeling the search space as a complete undirected graph, where the nodes represent features, and the weights of the edges represent the similarities between features. Each node in the graph has a *desirability* value called pheromone, which is updated by agents (ants). The agents traverse the graph iteratively preferring high pheromone values and low similarities until a pre-specified stop criterion (number of iterations) is reached. In the end, those features with the highest pheromone value are selected.

MGSACO: Microarray Gene Selection based on Ant Colony Optimization (Tabakhi et al., 2015) combines the computational efficiency of the filter approach and the good performance of the ACO approach. Similar to UFSACO, MGSACO represents the search space as a fully connected weighted graph, but with the difference that MGSACO proposes a fitness function based on the relevance and redundancy analyses (variance and cosine as relevance and similarity functions respectively) to identify irrelevant and redundant features. The search process stops until a maximum number of iterations is reached, and the global best subset of features in all iterations is chosen as the final subset of features.

RRFSACO: Relevance-Redundancy Feature Selection based on ACO (Tabakhi & Moradi, 2015) is a bio-inspired filter method that selects features in three steps: (i) initialization step, (ii) feature probability computation step, and (iii) final feature subset selection step. In the first step, the relevance of each feature and the similarity values between each pair of features are evaluated by the variance and cosine similarity function, respectively. The second step is used to compute the pheromone values of each feature in an iterative process. Finally, in the third step, the features are sorted in decreasing order of their pheromone values and the top p features are kept as the final subset of features.

DSRMR: Dual Self-Representation and Manifold Regularization (Tang et al., 2018a) is a filter sparse learning method based on the self-representation of features (Zhu, Zuo, Zhang, Hu, & Shiu, 2015). The idea is to optimize a model (objective function), take into account three aspects: (1) the self-representation of features using the $l_{2,1}$-norm, (2) the local manifold geometrical structure of the original data using a graph-based norm regularization term, and (3) a regularization term W to reflect the importance of each feature. The optimization problem is solved through an efficient iterative optimization algorithm. At the final stage, each feature is sorted according to the corresponding W values in descending order and the top p ranked features are selected.

3. Experimental protocol

In this section, we describe the methodology followed in this study for evaluating the filter UFS methods considered in this work.

3.1. Datasets

For our study, we selected 50 datasets from the UCI Machine Learning Repository[3] (Lichman, 2013) and the 25 datasets with the highest dimensionality from the ASU Feature Selection Repository[4] (Li et al., 2016). All of them are real-world datasets from different sources, which include text, face images, biological data, among others. Detailed information about the selected datasets is summarized in Tables 1–2.

Because some datasets have ranges of values with different scales for certain numerical features (which could affect the outcome of feature selection methods, clustering, and classification algorithms), all datasets in Tables 1–2 were standardized. That is, each dimension was normalized to obtain mean 0 and standard deviation 1, as recommended in Dy and Brodley (2004). Moreover, the missing values for non-numerical and numerical features were replaced by the modes and means, respectively. For all datasets, class labels were removed for feature selection and clustering. Furthermore, as usual, for the feature selection methods that can only process numerical features, the non-numerical features were transformed into numerical ones by mapping each categorical value into an integer value in the order of appearance of the dataset.

3.2. Performance evaluation

For our experiments, we follow the standard ways to assess Unsupervised Feature Selection methods (Li et al., 2016), i.e., in terms of clustering (Zhu et al., 2015; Zhu, Zhu, Wang, Zuo, & Hu, 2017; Zhou, Wu, Yi, & Luo, 2017; Wang & Wang, 2017; Tang et al., 2018b) and classification (Zhao & Liu, 2007; Padungweang, Lursinsap, & Sunat, 2009; Niijima & Okuno, 2009; Tabakhi et al., 2014; Tabakhi & Moradi, 2015; Tabakhi et al., 2015; Dadaneh, Markid, & Zakerolhosseini, 2016) performance. Additionally, we evaluate the runtime spent by each filter UFS method for performing feature selection.

For evaluating in terms of clustering performance, each feature selection method is first applied over the original dataset to select/rank features. Then, the well-known k-means clustering algorithm[5] (MacQueen, 1967) is applied over the data described just by the selected features. For evaluating the clustering results, the commonly used clustering performance metrics ACC (clustering ACCuracy) and the NMI (Normalized Mutual Information) were applied. The ACC and NMI evaluation measures are defined as follows:

Denoting by q_i the label computed by a clustering algorithm and by p_i the true label of an object $\boldsymbol{x}_i, i = 1, 2, \ldots, m$. *ACC* (He et al., 2005) is computed as follows:

$$ACC = \frac{\sum_{i=1}^{m} \delta(p_i, map(q_i))}{m} \tag{1}$$

where m is the total number of objects and $\delta(x, y) = 1$ if $x = y$; otherwise $\delta(x, y) = 0$. $map(q_i)$ is a mapping function that permutes clustering labels to get the best match with the true labels using the Kuhn-Munkres[6] algorithm (Lovász & Plummer, 1986). The *ACC* values range from 0 to 1, where $ACC \approx 1$ means a better clustering.

Given the true labels and the clustering results P and Q, respectively, NMI (He et al., 2005) is defined as:

$$NMI = \frac{I(P, Q)}{\max\{H(P), H(Q)\}} \tag{2}$$

where $H(P)$ and $H(Q)$ are the entropies of P and Q respectively, and $I(P, Q)$ is the mutual information (Cover & Thomas, 2006) between P and Q defined as follows:

$$I(P, Q) = \sum_{p_i \in P, q_j \in Q} p(p_i, q_j) \cdot \log_2 \frac{p(p_i, q_j)}{p(p_i) \cdot p(q_j)}$$

where $p(p_i)$ and $p(q_j)$ are the probabilities that an object arbitrarily selected from the dataset belongs to the clusters p_i and q_j, respectively, and $p(p_i, q_j)$ is the joint probability that the arbitrarily selected object belongs to the clusters p_i and q_j at the same time. Values of NMI range from 0 to 1. NMI reflects how similar are P and Q. Hence, a better clustering gets a larger NMI value.

[3] https://archive.ics.uci.edu/ml/index.php.

[4] http://featureselection.asu.edu/datasets.php.

[5] In order to get more reliable results, we repeat the k-means algorithm ten times with different initial points and report the average clustering quality results.

[6] http://www.cad.zju.edu.cn/home/dengcai/Data/code/hungarian.m.

Table 1
Description of the datasets taken from the UCI Machine Learning Repository.

#	Dataset	No. of Objects	No. of Features	No. of Classes
1	Inflamations	120	6	2
2	Audiology	226	69	24
3	Automovile	205	25	6
4	Breast-Cancer	286	9	2
5	Clean-2	6597	166	2
6	Contraception	1473	9	3
7	Credit-aproval	690	15	2
8	Credit-german	1000	20	2
9	Cylinder-bands	540	39	2
10	Dermatology	366	34	6
11	Ecoli	336	7	8
12	Flags	194	29	8
13	Heart-c	303	13	2
14	Heart-h	294	13	2
15	Heart-statlog	270	13	2
16	Hepatitis	155	19	2
17	Horse-colic	368	22	2
18	Hypothyroid	3772	29	4
19	Ionosphere	351	34	2
20	Iris	150	4	3
21	Nursery	12960	8	5
22	Liver-disorders	345	6	2
23	Lung-cancer	32	56	3
24	Lymphography	148	18	4
25	Promoters	106	58	2
26	Monks-1-train	124	6	2
27	Monks-2-train	169	6	2
28	Monks-3-train	122	6	2
29	Mushroom	8124	22	2
30	Optdigits	5620	64	10
31	Parkinsons	194	22	2
32	Pendigits	10992	16	10
33	Pima	768	8	2
34	Post-operative	90	8	3
35	Segment	2310	19	7
36	Sonar	208	60	2
37	Soybean	683	35	19
38	Spambase	4601	57	2
39	Splice	3190	61	3
40	Sponge	76	45	3
41	Tae	151	5	3
42	Thoracic	470	16	2
43	Tic-tac-toe	958	9	2
44	Silhouettes	846	18	4
45	Vote	435	16	2
46	Vowel	990	13	11
47	Waveform-5000	5000	40	3
48	Wdbc	569	30	2
49	Wine	178	13	3
50	Zoo	101	17	7

On the other hand, for evaluating the filter UFS methods in terms of classification performance, we used three well-know classifiers: the k-Nearest Neighbor (kNN) (Fix & Hodges, 1951) using $k = 3$, Support Vector Machine (SVM) (Cortes & Vapnik, 1995), and Naive Bayes (NB) (Maron, 1961; John & Langley, 1995). These three classifiers were chosen because they are among the most used classifiers for validating UFS methods. For the evaluation, because several datasets have few objects, we applied stratified 5-fold cross-validation, and the final classification performance (Classification accuracy) is reported as the average over the five folds. For each fold, each UFS method is first applied on the training set (ignoring the class labels), to obtain a feature subset. Then, after training the classifier using the selected features, the respective test sets are used for assessing each classifier through its accuracy.

In our experiments, for UCI datasets, in order to make a fair comparison among the different filter UFS methods analyzed in this paper, and because the optimal number of selected features is unknown, we set different percentages[7] of selected features for all datasets, and the best classification and clustering results in the different percents were reported as the final result for each feature selection method. Meanwhile, for ASU datasets, we used the 500 top-ranked features for those UFS methods that provide a feature ranking as output, and the same number of the whole set of features for those that need, as an input parameter, the number of features to select.

3.3. Parameter setting and used implementations

Most of the implementations, source code, of the state-of-the-art feature selection methods analyzed in this study were taken

[7] We performed the experiment taking 30%, 40%, 50% and 60% of the ranked features for those UFS methods that provide a feature ranking as output, and the same percentages of the whole set of features for those that need the number of features to select as an input parameter.

Table 2
Description of the high dimensional datasets taken from the ASU Feature Selection Repository.

#	Dataset	No. of Objects	No. of Features	No. of Classes
1	Isolet	1560	617	26
2	Yale	165	1024	15
3	OLR	400	1024	40
4	WarpAR10P	130	2400	10
5	Colon	62	2000	2
6	WarpPIE10P	2010	2420	10
7	Lung	203	3312	5
8	COIL20	1440	1024	20
9	Lymphoma	96	4026	9
10	GLIOMA	50	4434	4
11	ALLAML	72	7129	2
12	Prostate_GE	102	5966	2
13	TOX_171	171	5748	4
14	Leukemia	72	7070	2
15	Nci9	69	9712	9
16	Carcinom	174	9182	11
17	Arcene	200	10000	2
18	Orlraws10P	100	10304	10
19	Pixraw10P	100	10000	10
20	RELATHE	1427	4322	2
21	PCMAC	1943	3289	2
22	BASEHOCK	1993	4862	2
23	CLL_SUB_111	111	11340	3
24	GLI_85	85	22283	2
25	SMK_CAN_187	187	19993	2

from the Matlab Feature Selection package[8] available in the ASU Feature Selection repository (Li et al., 2016) and the Kurdistan Feature Selection Tool,[9] using the parameters' values recommended by their respective authors. Some other implementations of the tested methods were downloaded from the corresponding author's home pages or they were directly provided by the authors. Meanwhile, the classifiers and the *k*-means clustering algorithm used in our experiments were taken from the Weka data mining software tool (Hall et al., 2009), using their default parameter values.

The parameter *k* indicating the number of clusters required by *k*-means and some UFS methods was set as the number of classes declared for each dataset. For numerical datasets, the object's similarity matrix required for building the Laplacian graph in SPEC, MCFS, NDFS, Laplacian Score, and USFSM methods was computed using the popular Gaussian radial basis function (Buhmann, 2003).

All experiments were run in Matlab® R2018a with Java 1.8, using a computer with an Intel Core i7-2600 3.40 GHz$\times$8 processor with 32 GB DDR4 RAM, running 64-bit Ubuntu 16.04 LTS (GNU/Linux 4.13.0-38 generic) operating system.

3.4. Statistical tests for performance comparison

In order to perform a comprehensive comparison of the analyzed filter UFS methods, the Friedman test (Friedman, 1937) and the Holm post hoc procedure (Holm, 1979) were employed to measure the statistical significance of the results. The Friedman test is a non-parametric test used to measure the statistical differences among several methods over multiple datasets. For each dataset, the evaluated methods are ranked separately based on their final quality results (classification or clustering performance). The method with the highest quality gets rank 1, the second-highest result gets rank 2, and so on. The null hypothesis in the Friedman test means that all methods perform equivalently at the significance level α. In our experiments, the significance level was set to 0.05. If statistical differences are detected, the Holm post hoc test is applied to compare every pair of UFS methods. In the Holm test, if certain null hypothesis *i* cannot be rejected, all the remaining hypotheses will not be rejected as well. The statistical tests were performed using the Non-Parametric Statistical Analysis module of KEEL[10] (Triguero et al., 2017) and the statistical module of the Kurdistan Feature Selection Tool.[11]

4. Evaluation and analysis

In this section, we present the results of the systematic evaluation of the filter UFS methods considered in our study. Afterward, we give a general discussion pointing out some important facts derived from our experiments.

4.1. Experimental results

First, we will present the results of filter UFS methods in terms of clustering using *k*-means. Then, we will expose the results in terms of classification using SVM, *k*NN, and NB. And finally, we will show the results in terms of runtime.

4.1.1. Results in terms of clustering

The results of the evaluated filter UFS methods, in terms of ACC over the 50 datasets of Table 1, appear in Table 3, while the results in terms of NMI are shown in Table 4. The same results but for the 25 datasets of Table 2 are shown in Tables 5–6 respectively. In all these tables, the rows correspond to the datasets while the columns correspond to the filter UFS methods (univariate and multivariate). The best result for each dataset appears in "**bold**", and the last row shows the average rank obtained by each UFS method over all tested datasets. The values marked with the symbol "†" indicate that for the UFS method in the respective column it was not possible to obtain a result in the dataset of the respective row due to runtime errors in the authors' code or because they were not able to finish the feature selection after a period of time; thus, for comparison purposes, the value that appears in this entry is the average value of those methods that actually completed the feature selec-

8 http://featureselection.asu.edu/algorithms.php.
9 https://kfst.uok.ac.ir/relatedalgos.html.
10 http://www.keel.es/.
11 https://kfst.uok.ac.ir/.

Table 3
ACC results of filter UFS methods on the UCI datasets of Table 1 using *k*-means.

Dataset	*Univariate*						*Multivariate*												
	Variance	SUD	LS	SVD-Entropy	SPEC	USFSM	FSFS	MC	MCFS	MRSF	UDFS	NDFS	RRFS	JELSR	UFSACO	MGSACO	RRFSACO	DSRMR	**Original**
Inflammations	0.833	0.848	0.815	0.918	0.815	0.775	0.848	0.738	0.815	0.918	0.758	**1.000**	0.833	0.723	0.745	0.833	0.745	0.725	0.768
Audiology	0.352	0.331	0.350	**0.419**	0.310	0.354	0.342	0.323	0.395	0.358	0.352	0.391	0.418	0.341	0.352	0.342	0.363	0.358	0.337
Automovile	0.368	0.397	0.401	0.402	0.382	**0.433**	0.382	0.385	0.391	0.413	0.403	0.377	0.353	0.427	0.392	0.387	0.366	0.366	0.401
Breast-cancer	0.651	0.643	0.595	0.629	0.616	0.659	0.654	0.594	0.562	0.595	**0.726**	0.564	0.654	0.594	0.629	0.651	0.708	0.563	0.609
Contraception	0.397	0.391	0.398	0.400	0.379	0.394	0.407	0.398	0.400	0.398	**0.410**	0.407	0.402	**0.410**	0.384	0.397	0.383	0.398	0.388
Credit-approval	0.598	0.770	0.750	0.778	0.794	0.737	0.801	0.553	0.797	0.794	0.682	0.806	**0.820**	0.771	0.575	0.597	0.575	0.717	0.751
Credit-german	0.579	0.547	0.555	0.542	0.589	0.539	0.582	0.594	0.527	0.535	0.582	0.568	0.586	0.535	**0.646**	0.568	**0.646**	0.601	0.563
Cylinder-bands	**0.588**	0.574	0.560	0.564	0.542	0.558	0.553	0.527	0.565	0.550	0.561	0.538	0.569	0.564	0.534	0.542	0.519	0.569	0.551
Dermatology	0.673	0.561	0.648	0.663	0.632	0.631	0.523	0.422	0.679	**0.703**	0.658	0.634	0.632	0.694	0.649	0.679	0.656	0.615	0.668
Ecoli	0.570	0.443	0.524	0.467	0.439	0.450	0.487	0.524	0.467	0.512	0.422	0.467	0.482	0.465	0.499	0.465	**0.587**	0.443	**0.587**
Flags	**0.407**	0.307	0.320	0.370	0.315	0.335	0.326	0.298	0.345	0.346	0.318	0.340	0.319	0.330	0.334	0.345	0.281	0.363	0.344
Heart-c	0.782	0.828	0.818	0.785	0.786	**0.838**	0.683	0.751	0.802	0.724	0.791	0.771	0.795	0.719	0.779	0.792	0.779	0.736	0.809
Heart-h	0.830	0.813	0.790	0.792	0.813	0.830	0.667	0.624	0.814	0.790	0.814	0.833	**0.840**	0.813	0.691	0.704	0.626	0.728	0.788
Heart-statlog	0.739	0.763	0.768	0.789	0.763	0.776	0.727	0.619	0.762	0.762	0.739	0.789	**0.796**	0.758	0.679	0.727	0.679	0.763	0.796
Hepatitis	0.752	0.757	0.725	0.746	**0.768**	0.757	0.726	0.628	0.702	0.737	0.671	0.677	0.701	0.733	0.697	0.672	0.690	0.761	0.675
Horse-colic	0.583	0.583	**0.690**	0.546	0.575	0.564	0.680	0.623	0.601	0.592	0.552	0.604	0.648	0.635	0.634	0.576	0.642	0.604	0.559
Hypothyroid	0.458	0.355	0.817	0.529	0.463	0.466	0.405	**0.932**	0.709	0.888	0.522	0.458	0.464	0.749	0.838	0.555	0.880	0.617†	0.458
Ionosphere	0.699	0.715	0.707	0.707	0.698	0.707	**0.726**	0.701	0.704	0.698	0.712	0.712	0.682	0.709	0.717	0.707	0.693	0.716	0.711
Iris	**0.960**	**0.960**	**0.960**	**0.960**	**0.960**	**0.960**	0.917	0.795	0.835	0.917	0.917	0.835	0.917	**0.960**	0.795	**0.960**	0.795	0.795	0.883
Liver-disorders	0.553	0.559	0.556	0.559	0.559	0.559	0.553	**0.578**	0.546	0.561	0.547	0.559	0.556	0.546	**0.578**	0.553	**0.578**	0.559	0.548
Lung-cancer	0.575	0.550	0.575	0.563	0.506	**0.644**	0.556	0.469	0.613	0.538	0.600	0.563	0.500	0.606	0.588	0.538	0.538	0.575	0.594
Lymphography	0.457	**0.559**	0.484	0.546	0.492	0.501	0.486	0.482	0.532	0.557	0.497	0.482	0.380	0.557	0.518	0.474	0.500	0.557	0.458
Promoters	0.628	0.598	0.685	**0.709**	0.585	0.687	0.568	0.611	0.617	0.672	0.653	0.643	0.509	0.623	0.653	0.643	0.634	0.566	0.617
Monks-1-train	0.602	0.627	0.600	0.597	0.579	0.579	0.587	0.535	0.600	0.576	0.566	0.627	**0.674**	0.576	0.627	0.627	0.627	0.569	0.577
Monks-2-train	0.579	0.550	**0.598**	0.559	0.546	0.538	0.538	0.566	**0.598**	**0.598**	0.541	0.579	0.568	**0.598**	**0.598**	0.566	0.551	**0.598**	0.543
Monks-3-train	0.566	0.634	0.587	0.625	0.605	0.605	0.634	0.587	0.564	0.564	0.605	0.605	0.590	0.605	0.608	0.587	0.605	**0.636**	0.561
Parkinsons	0.695	0.577	0.739	0.603	0.704	0.588	0.725	0.754	0.746	0.726	0.704	0.745	0.745	0.696	0.740	0.729	**0.780**	0.644	0.629
Pima	0.664	0.666	0.663	0.669	0.665	0.671	0.671	0.643	0.694	0.672	0.622	0.671	0.659	0.663	0.671	0.664	0.671	**0.732**	0.668
Post-Operative	0.476	0.529	0.469	0.467	0.518	0.429	0.469	0.487	0.480	0.436	**0.613**	0.489	0.489	0.438	0.462	0.476	0.489	**0.613**	0.420
Segment	0.714	0.563	0.604	0.604	0.632	0.673	0.534	0.605	0.696	**0.742**	0.604	0.736	0.331	0.604	0.555	0.618	0.555	0.610†	0.606
Sonar	0.555	0.570	0.577	0.554	0.596	0.539	0.587	0.588	0.574	0.536	0.544	0.527	0.620	0.601	**0.625**	0.580	0.598	0.545	0.549
Soybean	0.437	0.600	0.634	0.615	0.616	0.522	0.386	0.373	0.591	0.548	0.620	**0.649**	0.414	0.608	0.542	0.518	0.549	0.576	0.472
Sponge	0.474	0.487	0.497	0.474	0.458	0.497	0.492	0.492	0.492	0.461	0.534	0.463	**0.774**	0.655	0.532	0.526	0.563	0.500	0.484
Tae	0.384	0.384	**0.465**	0.409	0.454	**0.465**	**0.465**	0.415	**0.465**	0.408	**0.465**	0.454	0.392	0.413	0.449	0.384	0.449	0.384	0.423
Thoracic	0.582	0.760	0.820	0.679	0.679	0.569	0.582	0.702	0.805	0.805	0.677	0.681	0.556	0.734	0.617	0.575	0.710	**0.826**	0.581
Tic-tac-toe	0.534	0.548	0.571	0.546	0.573	0.571	0.577	0.559	**0.593**	0.572	0.571	0.558	0.555	0.568	0.546	0.548	0.561	0.558	0.538
Silhouettes	0.382	**0.449**	0.444	**0.449**	0.377	0.441	0.383	0.374	0.372	0.381	0.439	0.390	0.372	0.405	0.364	0.387	0.377	0.428	0.370
Vote	0.841	0.885	0.876	0.885	0.839	0.881	0.800	0.789	0.854	0.877	0.901	0.894	0.553	0.886	**0.903**	0.879	**0.903**	0.793	0.869
Vowel	0.207	0.192	0.134	0.133	**0.274**	0.127	0.247	0.246	0.205	0.263	0.133	0.211	0.189	0.125	0.160	0.202	0.156	0.168	0.123
Wdbc	0.925	0.928	0.931	0.928	0.940	0.930	0.953	0.936	0.946	0.942	0.924	0.927	**0.958**	0.926	0.938	0.946	0.943	0.903	0.928
Wine	0.938	0.853	0.911	0.957	0.947	0.957	0.916	0.866	0.944	0.949	0.957	**0.973**	0.935	0.943	0.943	0.925	0.943	0.942	0.949
Zoo	0.590	0.885	0.885	0.893	0.697	0.644	0.721	0.562	0.784	0.723	**0.895**	0.719	0.513	0.850	0.806	0.776	0.804	0.816	0.707
Splice	0.392	0.375	0.410	0.418	0.380	0.428	0.374	0.374	0.388	0.389	0.398	0.397	**0.519**	0.415	0.397	0.392	0.398	0.379	0.409
Spambase	0.767	0.609	0.676	0.714	0.617	0.685	0.791	0.613	0.717	0.730	0.701	0.690	**0.791**	0.728	0.622	0.682	0.659	0.694†	0.675
Mushroom	0.747	0.788	0.661	0.673	0.654	**0.891**	0.764	0.666	0.659	0.770	0.709	0.692	0.553	0.730	0.693	0.689	0.750	0.711†	0.701
Optdigits	0.754	0.602	0.718	0.722	0.668	0.753	0.744	0.621	0.713	0.751	0.681	0.640	**0.762**	0.659	0.702	0.657	0.662	0.695†	0.739
Waveform-5000	0.522	0.511	0.516	0.516	0.446	0.516	0.478	0.474	0.518	0.513	0.521	0.512	**0.522**	0.516	0.519	0.518	0.520	0.508†	0.512
Pendigits	0.621	0.633	0.666	0.683	0.587	0.691	0.614	0.694	0.668	0.672	0.674	0.691	0.694	0.691	0.654	0.602	0.648	0.658†	**0.718**
Clean2	0.541	0.540	0.541	0.540	0.544	0.540	0.548	0.560	0.536	0.539	0.537	0.541	0.543	0.541	**0.581**	0.544	0.581	0.547†	0.538
Nursery	0.293	0.351†	**0.422**	0.295	**0.422**	0.351†	**0.442**	0.293	0.419	0.420	**0.422**	0.293	0.298	0.351†	0.301	0.297	0.293	0.351†	0.351
Average Rank	**10.17**	**9.97**	**8.52**	**8.48**	**10.78**	**8.7**	**9.84**	**12.45**	**8.54**	**8.6**	**9.15**	**8.87**	**9.2**	**8.78**	**9.17**	**10.73**	**9.41**	**9.64**	–

Table 4
NMI results of filter UFS methods on the UCI datasets of Table 1 using *k*-means.

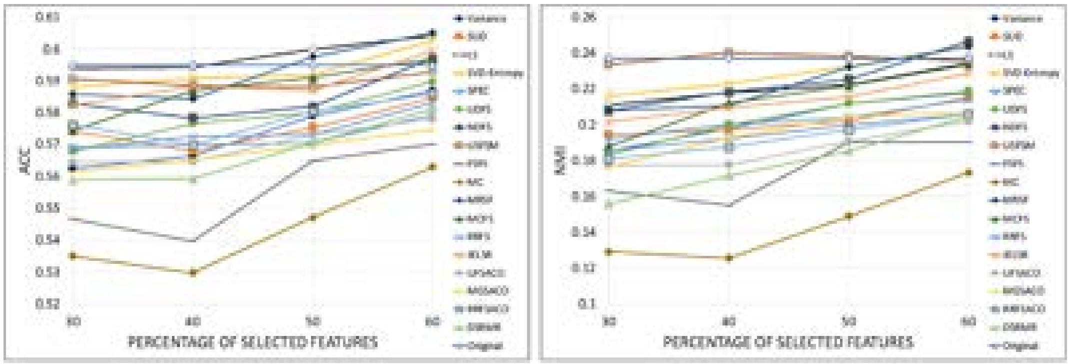
Dataset	*Univariate*						*Multivariate*												
	Variance	SUD	LS	SVD-Entropy	SPEC	USFSM	FSFS	MC	MCFS	MRSF	UDFS	NDFS	RRFS	JELSR	UFSACO	MGSACO	RRFSACO	DSRMR	**Original**
Inflammations	0.379	0.462	0.440	0.756	0.440	0.374	0.462	0.287	0.440	0.756	0.289	**1.000**	0.350	0.284	0.286	0.379	0.286	0.274	0.366
Audiology	0.472	0.217	0.469	0.534	0.410	0.472	0.476	0.247	0.504	0.472	0.364	0.528	**0.550**	0.464	0.481	0.471	0.502	0.310	0.460
Automovile	0.175	0.201	0.183	0.199	0.181	0.265	0.249	0.234	0.217	0.246	0.257	0.169	0.183	**0.281**	0.245	0.206	0.231	0.171	0.244
Breast-cancer	0.012	0.019	0.010	0.024	0.028	0.021	0.006	0.003	0.010	0.010	0.051	0.006	0.007	0.012	0.027	0.020	**0.051**	0.004	0.021
Contraception	0.023	0.028	0.015	0.026	0.014	0.023	**0.029**	0.017	0.018	0.021	0.021	0.019	0.023	0.021	0.024	0.024	0.024	0.015	0.022
Credit-approval	0.024	0.268	0.220	0.272	0.266	0.238	0.287	0.011	0.300	0.300	0.142	0.291	**0.345**	0.238	0.016	0.025	0.016	0.210	0.216
Credit-german	0.017	0.014	0.004	0.006	0.003	0.003	0.023	0.016	0.003	0.003	0.014	0.006	0.006	0.002	**0.040**	0.017	0.039	0.024	0.013
Cylinder-bands	**0.026**	0.015	0.009	0.010	0.013	0.008	0.007	0.001	0.012	0.005	0.010	0.007	0.017	0.010	0.014	0.018	0.001	0.012	0.006
Dermatology	0.606	0.536	0.649	0.664	0.602	0.617	0.450	0.256	0.674	0.677	0.602	0.599	0.580	**0.678**	0.627	0.646	0.620	0.570	0.616
Ecoli	0.489	0.415	0.484	0.473	0.418	0.438	0.433	0.484	0.473	0.454	0.321	0.473	0.455	0.452	0.477	0.452	**0.535**	0.415	**0.535**
Flags	**0.268**	0.152	0.181	0.243	0.155	0.217	0.164	0.158	0.226	0.218	0.152	0.220	0.153	0.204	0.195	0.196	0.150	0.241	0.213
Heart-c	0.247	0.335	0.313	0.246	0.256	**0.358**	0.117	0.190	0.278	0.148	0.279	0.268	0.268	0.143	0.240	0.285	0.240	0.202	0.292
Heart-h	0.311	0.264	0.253	0.277	0.264	0.311	0.090	0.028	0.302	0.253	0.302	0.320	**0.331**	0.264	0.081	0.176	0.059	0.154	0.250
Heart-statlog	0.190	0.205	0.220	0.254	0.205	0.230	0.172	0.065	0.209	0.209	0.190	0.254	**0.270**	0.201	0.105	0.170	0.105	0.205	0.270
Hepatitis	**0.204**	0.133	0.106	0.139	0.192	0.154	0.106	0.051	0.086	0.143	0.072	0.087	0.107	0.147	0.043	0.102	0.092	0.177	0.107
Horse-colic	0.060	0.069	0.134	0.046	0.052	0.058	**0.154**	0.115	0.085	0.073	0.044	0.084	0.116	0.099	0.116	0.052	0.116	0.079	0.057
Hypothyroid	0.005	0.017	0.010	0.019	0.007	0.006	0.005	**0.274**	0.008	0.161	0.035	0.006	0.006	0.011	0.011	0.017	0.169	0.045†	0.005
Ionosphere	0.113	0.125	0.126	0.126	0.108	0.126	**0.183**	0.147	0.119	0.106	0.131	0.131	0.082	0.128	0.132	0.110	0.101	0.130	0.128
Iris	**0.867**	0.864	0.864	0.864	**0.867**	0.864	0.805	0.601	0.663	0.805	0.805	0.663	0.805	0.864	0.601	**0.867**	0.601	0.601	0.723
Liver-disorders	0.001	0.001	0.002	0.001	0.003	0.003	0.001	**0.010**	0.001	0.002	0.001	0.001	0.002	0.001	**0.010**	0.001	**0.010**	0.001	0.000
Lung-cancer	0.273	0.258	0.292	0.282	0.176	**0.385**	0.179	0.092	0.294	0.279	0.271	0.282	0.147	0.264	0.226	0.201	0.197	0.224	0.279
Lymphography	0.131	0.139	0.140	0.144	0.138	0.150	0.148	0.116	0.145	0.157	0.125	0.117	0.055	0.160	0.185	0.134	**0.196**	0.160	0.132
Promoters	0.079	0.030	0.154	**0.154**	0.027	0.137	0.019	0.053	0.075	0.120	0.081	0.087	0.009	0.063	0.088	0.098	0.079	0.019	0.066
Monks-1-train	0.045	0.068	0.036	0.045	0.030	0.030	0.069	0.006	0.036	0.026	0.027	0.068	**0.183**	0.026	0.068	0.068	0.068	0.024	0.030
Monks-2-train	0.007	0.002	**0.017**	0.002	0.008	0.005	0.004	0.009	**0.017**	**0.017**	0.012	0.007	0.008	**0.017**	**0.017**	0.009	0.003	**0.017**	0.006
Monks-3-train	0.019	**0.087**	0.058	0.058	0.073	0.073	**0.087**	0.061	0.052	0.052	0.042	0.073	0.060	0.042	0.043	0.061	0.073	0.073	0.023
Parkinsons	**0.230**	0.103	0.227	0.100	0.216	0.192	0.133	0.166	0.168	0.139	0.221	0.187	0.091	0.133	0.167	0.228	0.147	0.137	0.215
Pima	0.049	0.051	0.049	0.054	0.045	0.054	0.054	0.034	0.074	0.060	0.037	0.054	0.045	0.049	0.054	0.049	0.054	**0.131**	0.051
Post-Operative	0.026	0.020	0.017	**0.030**	0.017	0.019	0.015	0.019	0.026	0.019	0.023	0.026	0.026	0.021	0.021	0.026	0.025	0.022	0.023
Segment	0.671	0.575	0.618	0.618	0.631	0.665	0.542	0.565	0.668	0.697	0.618	**0.697**	0.247	0.612	0.527	0.600	0.527	0.593†	0.596
Sonar	0.009	0.030	0.019	0.007	**0.079**	0.005	0.019	0.022	0.020	0.008	0.007	0.002	0.039	0.029	0.042	0.020	0.028	0.012	0.007
Soybean	0.548	0.667	0.717	0.711	**0.724**	0.668	0.495	0.433	0.702	0.670	0.707	0.715	0.481	0.704	0.616	0.624	0.647	0.682	0.593
Sponge	0.053	0.053	0.052	0.052	0.050	0.057	0.060	0.047	0.050	0.047	0.053	0.046	0.066	**0.076**	0.052	0.049	0.050	0.038	0.041
Tae	0.024	0.024	**0.062**	0.057	0.059	**0.062**	0.060	0.047	**0.062**	0.057	**0.062**	0.059	0.026	0.035	0.055	0.024	0.055	0.024	0.044
Thoracic	0.003	0.006	0.009	0.004	0.005	0.003	0.003	0.005	0.009	0.009	0.005	0.005	0.001	0.009	0.003	0.003	0.004	**0.010**	0.002
Tic-tac-toe	0.003	0.003	0.010	0.003	0.005	0.010	0.011	0.006	**0.025**	0.017	0.010	0.012	0.006	0.010	0.003	0.003	0.007	0.005	0.001
Silhouettes	0.103	**0.182**	**0.182**	**0.182**	0.124	0.167	0.121	0.104	0.122	0.105	0.177	0.129	0.111	0.145	0.115	0.092	0.118	0.154	0.118
Vote	0.373	0.490	0.476	0.490	0.401	0.478	0.350	0.314	0.466	0.502	0.526	0.517	0.032	0.506	**0.549**	0.477	**0.549**	0.339	0.465
Vowel	0.120	0.087	0.054	0.052	0.268	0.041	0.233	0.252	0.203	**0.288**	0.052	0.242	0.085	0.042	0.062	0.110	0.052	0.069	0.042
Wdbc	0.600	0.625	0.628	0.625	0.660	0.619	0.719	0.670	0.693	0.689	0.611	0.605	**0.762**	0.615	0.686	0.690	0.704	0.538	0.611
Wine	0.774	0.634	0.728	0.846	0.800	0.846	0.697	0.583	0.798	0.823	0.835	**0.889**	0.800	0.808	0.798	0.757	0.798	0.815	0.835
Zoo	0.643	0.881	0.882	0.882	0.667	0.646	0.748	0.492	0.790	0.740	**0.901**	0.724	0.425	0.847	0.803	0.761	0.778	0.817	0.752
Splice	0.015	0.006	0.024	0.029	0.018	**0.039**	0.006	0.008	0.020	0.018	0.023	0.026	0.007	0.024	0.020	0.019	0.020	0.015	0.032
Spambase	0.198	0.011	0.109	0.147	0.036	0.155	0.241	0.020	0.136	0.159	0.123	0.112	**0.241**	0.155	0.048	0.102	0.096	0.123†	0.110
Mushroom	0.231	0.327	0.102	0.123	0.089	**0.575**	0.248	0.148	0.127	0.328	0.199	0.134	0.034	0.180	0.132	0.140	0.254	0.198†	0.219
Optdigits	0.705	0.551	0.677	0.682	0.654	0.704	0.681	0.588	0.690	0.713	0.664	0.626	0.702	0.642	0.637	0.609	0.637	0.657†	**0.715**
Waveform-5000	0.362	0.365	0.365	0.365	0.157	0.365	0.099	0.171	0.363	0.364	0.363	0.365	0.361	0.365	0.355	**0.365**	0.355	0.324†	0.364
Pendigits	0.580	0.620	0.649	0.646	0.593	0.666	0.569	0.651	0.657	0.647	0.668	0.666	0.653	0.666	0.589	0.564	0.592	0.628†	**0.678**
Clean2	0.020	0.024	0.023	0.020	0.016	0.024	0.015	0.019	0.024	0.020	0.024	0.020	0.019	0.020	0.028	0.023	**0.028**	0.022†	0.024
Nursery	0.017	0.087†	0.171	0.017	0.171	0.087†	0.146	0.017	0.158	**0.191**	0.171	0.017	0.085	0.087†	0.024	0.015	0.017	0.087†	0.052
Average Rank	**10.13**	**9.77**	**8.32**	**8.1**	**10.53**	**7.57**	**10.36**	**12.91**	**7.85**	**8.12**	**9.13**	**8.65**	**10.45**	**8.9**	**9.47**	**10.3**	**9.72**	**10.72**	–

Table 5
ACC results of filter UFS methods on the ASU datasets of Table 2 using *k*-means.

Dataset	*Univariate*						*Multivariate*												
	Variance	SUD	LS	SVD-Entropy	SPEC	USFSM	FSFS	MC	MCFS	MRSF	UDFS	NDFS	RRFS	JELSR	UFSACO	MGSACO	RRFSACO	DSRMR	**Original**
Yale	0.393	0.417	0.435	**0.444**	0.392	0.379	0.365	0.373	0.413	0.402	0.390	0.400	0.389	0.348	0.393	0.401	0.402	0.382	0.401
Isolet	0.549	0.526	0.559	0.526	**0.647**	0.554	0.556	0.592	0.558	0.597	0.542	0.603	0.585	0.570	0.573	0.546	0.557	0.505	0.581
ORL	0.502	0.489	0.498	0.490	0.485	0.514	0.485	0.503	**0.518**	0.512	0.485	0.507	0.485	0.498	0.502	0.513	0.500	0.488	0.500
WarpPIE10P	0.295	0.256	0.328	0.261	0.363	0.277	0.289	**0.388**	0.286	0.325	0.292	0.295	0.295	0.360	0.305	0.276	0.308	0.227	0.302
Colon	0.548	0.584	0.584	0.581	0.548	0.513	**0.587**	0.523	0.529	0.529	0.552	0.548	0.529	0.523	0.519	0.529	0.532	0.539	0.548
WarpAR10P	0.206	0.240	0.231	0.242	0.308	0.258	0.272	**0.348**	0.223	0.251	0.217	0.232	0.306	0.275	0.214	0.229	0.209	0.246	0.246
Lung	0.708	0.618	**0.808**	0.648	0.694	0.629	0.589	0.527	0.634	0.606	0.696	0.794	0.609	0.540	0.755	0.698	0.714	0.626	0.725
COIL20	0.589	0.555	0.587	0.584	0.583	0.592	0.532	0.591	0.611	0.588	0.532	0.594	**0.613**	0.549	0.578	0.602	0.574	0.602	0.595
Lymphoma	0.488	0.504	0.510	0.504	0.435	0.502	0.350	0.485	**0.550**	0.533	0.531	0.513	0.502	0.425	0.506	**0.550**	0.523	0.498	0.538
GLIOMA	0.580	0.524	0.552	0.520	0.584	0.544	0.556	0.452	**0.660**	0.524	0.576	0.560	0.580	0.456	0.536	0.588	0.552	0.532	0.580
ALLAML	0.706	0.697	0.644	**0.731**	0.703	0.553	0.650	0.656	0.714	0.592	0.642	0.708	0.686	0.647	0.678	0.689	0.628	0.728	0.686
CLL_SUB_111	0.486	0.506†	**0.548**	0.506†	0.546	0.517	0.494	0.459	0.501	0.521	0.501	0.535	0.488	0.492	0.505	0.497	0.479	0.530	0.508
Carcinom	0.571	0.526†	0.534	0.526†	0.605	0.540	0.461	0.338	**0.667**	0.521	0.534	0.577	0.498	0.317	0.616	0.592	0.624	0.417	0.582
GLI_85	0.699	0.682†	0.687	0.682†	0.656	0.694	0.581	0.668	**0.720**	0.664	0.678	0.706	**0.720**	0.713	0.678	0.671	0.692	0.687	0.692
Prostate_GE	0.586	0.566†	0.569	0.566†	**0.596**	0.561	0.555	0.541	0.559	0.563	0.557	0.588	0.576	0.539	0.567	0.571	0.567	0.569	0.586
SMK_CAN_187	0.547	0.563†	0.562	0.563†	0.573	0.556	0.513	0.562	0.587	0.549	0.581	0.524	**0.639**	0.524	0.597	0.575	0.580	0.540	0.549
TOX_171	0.427	0.419†	0.483	0.419†	0.433	0.306	0.350	0.352	**0.538**	0.408	0.415	0.476	0.433	0.324	0.418	0.468	0.436	0.436	0.441
Arcene	0.626	0.617†	0.629	0.617†	0.640	0.556	0.637	0.558	0.620	0.643	0.564	**0.655**	0.635	0.551	0.637	0.648	0.628	0.650	0.643
Leukemia	0.689	0.666†	0.692	0.666†	0.664	0.611	0.625	0.661	0.706	0.597	0.669	**0.708**	0.692	0.675	0.664	0.675	0.664	0.661	0.675
Nci9	0.397	0.372†	0.377	0.372†	0.397	0.350	0.360	0.273	0.433	0.327	0.360	0.393	**0.440**	0.243	0.427	0.393	0.430	0.357	0.407
Orlraws10P	0.678	0.606†	**0.750**	0.606†	0.602	0.484	0.566	0.530	0.688	0.550	0.590	0.590	0.654	0.432	0.624	0.680	0.624	0.658	0.688
Pixraw10P	0.646	0.642†	**0.714**	0.642†	0.662	0.674	0.594	0.604	0.662	0.626	0.688	0.620	0.680	0.510	0.686	0.672	0.664	0.576	0.664
RELATHE	0.541	0.544†	0.540	0.544†	0.545	0.541	0.542	0.547	0.530	0.546	0.537	**0.550**	0.547	0.545	0.550	0.546	0.546	0.546	0.547
PCMAC	0.505	0.507†	0.505	0.507†	0.506	0.505	0.502	0.507	0.507	0.505	0.505	0.506	0.505	0.507	**0.518**	0.510	0.513	0.506	0.505
BASEHOCK	0.508	0.506†	0.505	0.506†	0.506	0.506	0.512	0.503	0.502	0.506†	0.506†	0.501	**0.513**	0.502	0.511	0.507	0.506	0.505	0.503
Average rank	**8.92**	**10.44**	**7.44**	**9.80**	**7.74**	**11.86**	**13.12**	**12.32**	**6.76**	**10.58**	**11.16**	**6.64**	**7.34**	**13.40**	**7.78**	**6.88**	**8.32**	**10.50**	-

Table 6
NMI results of filter UFS methods on the ASU datasets of Table 2 using k-means.

Dataset	*Univariate*						*Multivariate*												
	Variance	SUD	LS	SVD-Entropy	SPEC	USFSM	FSFS	MC	MCFS	MRSF	UDFS	NDFS	RRFS	JELSR	UFSACO	MGSACO	RRFSACO	DSRMR	**Original**
Yale	0.452	0.490	0.503	**0.521**	0.459	0.451	0.425	0.438	0.473	0.456	0.435	0.460	0.462	0.416	0.452	0.458	0.460	0.441	0.457
Isolet	0.710	0.694	0.725	0.702	**0.779**	0.718	0.693	0.722	0.724	0.754	0.710	0.757	0.731	0.722	0.723	0.705	0.717	0.689	0.736
ORL	0.719	0.707	0.714	0.703	0.691	**0.724**	0.689	0.710	0.723	0.722	0.698	0.714	0.710	0.706	0.710	0.720	0.710	0.705	0.717
WarpPIE10P	0.286	0.239	0.329	0.244	0.349	0.279	0.263	**0.433**	0.303	0.350	0.307	0.334	0.327	0.374	0.308	0.286	0.317	0.194	0.304
Colon	0.004	0.016	0.018	0.018	0.003	0.002	**0.078**	0.001	0.001	0.001	0.008	0.006	0.001	0.001	0.000	0.000	0.001	0.003	0.004
WarpAR10P	0.157	0.239	0.203	0.242	**0.351**	0.242	0.231	0.346	0.183	0.210	0.189	0.191	0.302	0.228	0.182	0.195	0.173	0.217	0.197
Lung	0.508	0.444	**0.575**	0.449	0.534	0.447	0.395	0.315	0.474	0.462	0.532	0.575	0.479	0.288	0.546	0.515	0.527	0.444	0.531
COIL20	0.740	0.716	0.716	0.730	0.690	0.738	0.655	0.721	**0.750**	0.743	0.693	0.742	0.749	0.694	0.732	0.739	0.733	0.739	0.745
Lymphoma	0.524	0.550	0.558	0.553	0.469	0.496	0.300	0.394	**0.601**	0.582	0.490	0.539	0.566	0.320	0.510	0.541	0.538	0.541	0.544
GLIOMA	0.459	0.440	0.475	0.444	0.496	0.373	0.296	0.250	**0.549**	0.421	0.476	0.502	0.433	0.226	0.412	0.454	0.408	0.477	0.495
ALLAML	0.161	0.095	0.135	0.142	0.108	0.091	0.054	0.056	**0.171**	0.018	0.096	0.122	0.090	0.045	0.106	0.170	0.064	0.140	0.134
CLL_SUB_111	0.166	0.162†	**0.239**	0.162†	0.206	0.199	0.121	0.016	0.197	0.111	0.123	0.203	0.151	0.108	0.164	0.173	0.174	0.210	0.185
Carcinom	0.598	0.532†	0.605	0.532†	0.607	0.545	0.447	0.225	**0.711**	0.496	0.515	0.638	0.543	0.221	0.615	0.604	0.643	0.423	0.614
GLI_85	0.112	0.077†	0.050	0.077†	0.081	0.054	0.006	0.030	0.143	0.035	0.018	**0.239**	0.169	0.037	0.024	0.030	0.105	0.078	0.105
Prostate_GE	0.023	0.018†	0.013	0.018†	0.029	0.014	0.009	0.017	0.013	**0.030**	0.012	0.025	0.017	0.022	0.017	0.020	0.017	0.013	0.023
SMK_CAN_187	0.006	0.017†	0.010	0.017†	0.017	0.013	0.002	0.036	0.023	0.007	0.019	0.001	**0.059**	0.023	0.029	0.024	0.019	0.004	0.007
TOX_171	0.196	0.178†	0.298	0.178†	0.197	0.020	0.066	0.083	**0.344**	0.108	0.139	0.285	0.180	0.058	0.159	0.232	0.211	0.227	0.228
Arcene	0.058	0.055†	0.057	0.055†	0.074	0.011	0.063	0.005	0.041	0.071	0.039	**0.086**	0.061	0.012	0.068	0.080	0.057	0.082	0.071
Leukemia	0.128	0.094†	0.108	0.094†	0.077	0.065	0.051	0.034	0.157	0.013	0.091	0.122	**0.183**	0.032	0.108	0.130	0.103	0.082	0.117
Nci9	0.394	0.350†	0.335	0.350†	0.372	0.325	0.359	0.196	0.417	0.277	0.339	0.374	**0.453**	0.164	0.434	0.377	0.445	0.312	0.378
Orlraws10P	0.763	0.697†	**0.828**	0.697†	0.689	0.591	0.640	0.586	0.769	0.640	0.691	0.713	0.770	0.474	0.714	0.754	0.716	0.739	0.766
Pixraw10P	0.763	0.765†	**0.850**	0.765†	0.773	0.798	0.714	0.713	0.801	0.755	0.818	0.739	0.775	0.602	0.813	0.810	0.793	0.694	0.793
RELATHE	0.001	0.002†	0.007	0.002†	0.001	0.001	0.000	0.001	**0.015**	0.001	0.001	0.003	0.001	0.002	0.001	0.001	0.001	0.000	0.002
PCMAC	0.000	0.001†	0.002	0.001†	0.001	0.003	0.000	0.001	**0.005**	0.000	0.001	0.000	0.000	0.002	0.003	0.001	0.002	0.000	0.001
BASEHOCK	0.002	0.002†	0.005	0.002†	0.002	0.002	0.001	0.002	0.001	0.002†	0.002†	0.001	0.003	0.003	0.001	0.002	0.001	**0.006**	0.001
Average rank	**8.74**	**9.98**	**6.12**	**9.26**	**7.86**	**10.82**	**14.70**	**12.70**	**5.72**	**10.52**	**11.48**	**6.60**	**7.04**	**12.92**	**9.18**	**7.78**	**8.94**	**10.64**	–



(a) ACC of k-means on UCI datasets (b) NMI of k-means on UCI datasets

Fig. 1. Average ACC and NMI over the results of the compared filter UFS methods using k-means, selecting the 30–60% of features of the UCI datasets of Table 1. The x-axis is the percentage of selected features, and the y-axis is the corresponding average clustering quality (ACC or NMI).

Table 7
Results of the Friedman test for the comparisons among filter UFS methods of Tables 3–6.

	Clustering algorithm	F_F	Critical value	Significant differences
UCI	k-means (ACC)	1.921	1.635	Yes
	k-means (NMI)	3.223	1.635	Yes
ASU	k-means (ACC)	5.251	1.647	Yes
	k-means (NMI)	6.566	1.647	Yes

Table 8
Results (p-values) achieved on post hoc comparisons among the results of the filter UFS methods of Tables 3–4 for $\alpha = 0.05$ using k-means.

i	UFS methods	p-value	Significant differences
k-means (ACC)			
1	SVD-Entropy vs. MC	0.000201	Yes
2	LS vs. MC	0.000233	Yes
3	MC vs. MCFS	0.00025	Yes
4	MC vs. MRSF	0.000311	Yes
5	USFSM vs. MC	0.000444	No
k-means (NMI)			
1	USFSM vs. MC	0.000001	Yes
2	MC vs. MCFS	0.000002	Yes
3	SVD-Entropy vs. MC	0.000007	Yes
4	MC vs. MRSF	0.000007	Yes
5	LS vs. MC	0.000017	Yes
6	MC vs. NDFS	0.000066	Yes
7	MC vs. JELSR	0.000173	Yes
8	MC vs. UDFS	0.0004	No

tion for this dataset. Moreover, to put in perspective the performance of the tested UFS methods in comparison to not performing feature selection, in the last column of these tables, we show the results obtained by using the whole set of features (original). Additionally, in order to see the behavior of the UFS methods for different percentages of selected features, in Fig. 1 we show the corresponding average results for 30%, 40%, 50% and 60% of the selected features for the datasets of Table 1.

The results of the Friedman test corresponding to the results of the filter UFS methods in terms of ACC and NMI over the UCI and ASU datasets are shown in Table 7. In Table 7, the first two rows correspond to the results for the UCI datasets, while the last two rows correspond to the results for the ASU datasets. The second column of Table 7 specifies the clustering algorithm along with the evaluation measure used, and the remaining three columns show the corresponding statistics Fisher distribution (F_F), critical values, and an indicator that shows if there are significant differences or not between the compared UFS methods for the results

Table 9
Results (p-values) achieved on post hoc comparisons among the results of the filter UFS methods of Tables 5–6 for $\alpha = 0.05$ using k-means.

i	UFS methods	p-value	Significant differences
k-means (ACC)			
1	NDFS vs. JELSR	0.000008	Yes
2	MCFS vs. JELSR	0.000011	Yes
3	JELSR vs. MGSACO	0.000016	Yes
4	FSFS vs. NDFS	0.000018	Yes
5	FSFS vs. MCFS	0.000025	Yes
6	FSFS vs. MGSACO	0.000036	Yes
7	RRFS vs. JELSR	0.00006	Yes
8	LS vs. JELSR	0.000079	Yes
9	FSFS vs. RRFS	0.000129	Yes
10	LS vs. FSFS	0.000169	Yes
11	MC vs. NDFS	0.000169	Yes
12	SPEC vs. JELSR	0.000178	Yes
13	JELSR vs. UFSACO	0.000198	Yes
14	MC vs. MCFS	0.000231	Yes
15	MC vs. MGSACO	0.000315	Yes
16	SPEC vs. FSFS	0.000367	No
k-means (NMI)			
1	FSFS vs. MCFS	0	Yes
2	LS vs. FSFS	0	Yes
3	FSFS vs. NDFS	0	Yes
4	FSFS vs. RRFS	0	Yes
5	MCFS vs. JELSR	0.000002	Yes
6	MC vs. MCFS	0.000004	Yes
7	FSFS vs. MGSACO	0.000005	Yes
8	SPEC vs. FSFS	0.000006	Yes
9	LS vs. JELSR	0.000007	Yes
10	LS vs. MC	0.000013	Yes
11	NDFS vs. JELSR	0.000028	Yes
12	MC vs. NDFS	0.000053	Yes
13	Variance vs. FSFS	0.000079	Yes
14	RRFS vs. JELSR	0.000099	Yes
15	MCFS vs. UDFS	0.000136	Yes
16	FSFS vs. RRFSACO	0.000136	Yes
17	MC vs. RRFS	0.000178	Yes
18	FSFS vs. UFSACO	0.000256	Yes
19	SVD-Entropy vs. FSFS	0.000315	Yes
20	LS vs. UDFS	0.000386	No

Table 10
Classification accuracy results of filter UFS methods on the UCI datasets of Table 1 using SVM.

Dataset	*Univariate*						*Multivariate*												
	Variance	SUD	LS	SVD-Entropy	SPEC	USFSM	FSFS	MC	MCFS	MRSF	UDFS	NDFS	RRFS	JELSR	UFSACO	MGSACO	RRFSACO	DSRMR	**Original**
Inflammations	0.842	**1.000**	**1.000**	0.958	**1.000**	**1.000**	0.942	0.842	0.942	0.933	0.925	**1.000**	0.833	0.925	0.875	0.858	0.917	0.925	**1.000**
Audiology	0.814	0.305	0.730	0.814	0.557	0.797	0.797	0.358	0.783	0.761	0.376	0.801	**0.836**	0.717	0.779	0.792	0.814	0.319	0.814
Automovile	0.546	0.634	0.498	0.498	0.478	0.673	0.659	0.668	0.688	0.639	0.507	0.571	0.668	0.502	0.673	0.654	0.673	0.561	**0.693**
Breast-cancer	0.689	0.696	0.724	0.710	0.703	0.692	0.717	0.685	0.724	**0.727**	0.706	0.717	0.692	0.720	0.710	0.696	0.699	0.699	0.685
Contraception	0.489	0.460	0.421	**0.500**	0.415	0.465	0.458	0.418	0.459	0.442	0.435	0.474	0.473	0.443	0.479	0.492	0.492	0.420	0.482
Credit-approval	0.745	**0.858**	0.852	**0.858**	0.854	0.852	0.857	0.745	0.852	0.851	0.797	0.855	0.830	0.852	0.742	0.751	0.749	0.752	0.848
Credit-german	0.753	0.756	0.709	0.717	0.700	0.741	0.732	0.726	0.706	0.728	0.729	0.735	0.706	0.722	0.733	0.748	0.730	0.752	**0.765**
Cylinder-bands	0.807	0.694	0.807	**0.817**	0.670	0.809	0.807	0.667	0.798	0.789	0.709	0.813	0.685	0.809	0.711	0.800	0.733	0.739	0.811
Dermatology	0.964	0.861	0.891	0.880	0.896	0.923	0.894	0.951	0.899	0.937	0.915	**0.975**	0.943	**0.975**	0.921	0.945	0.921	0.948	0.967
Ecoli	0.676	0.655	0.777	0.771	0.670	0.714	0.732	0.765	0.795	0.777	0.685	0.771	0.762	0.762	0.640	0.762	**0.831**	0.655	**0.831**
Flags	0.644	0.413	0.505	**0.649**	0.454	0.624	0.577	0.516	0.541	0.628	0.428	0.619	0.464	0.618	0.479	0.587	0.428	0.603	0.649
Heart-c	0.799	0.825	0.838	0.822	0.835	**0.848**	0.792	0.766	0.772	0.818	0.815	0.825	0.802	0.795	0.812	0.792	0.809	0.809	0.832
Heart-h	0.813	0.827	0.830	0.823	0.827	0.813	0.803	0.732	0.827	0.834	0.837	0.830	0.813	0.837	0.810	0.810	0.793	**0.840**	0.837
Heart-statlog	0.800	0.752	0.800	0.830	0.826	0.833	0.778	0.796	0.804	0.804	0.778	0.833	0.826	0.793	0.833	0.807	**0.837**	0.830	**0.837**
Hepatitis	0.839	0.852	0.845	0.852	**0.858**	0.826	0.839	**0.858**	0.845	0.845	**0.858**	0.852	0.826	0.845	0.845	0.832	0.845	0.845	0.845
Horse-colic	0.764	0.721	0.815	0.729	0.756	0.807	0.834	0.818	0.823	0.813	0.821	0.831	0.815	0.812	0.748	0.747	0.766	**0.848**	0.810
Hypothyroid	0.936	0.929	0.923	**0.936**	**0.936**	0.935	0.932	0.929	0.923	0.923	0.935	0.936	0.932	0.923	0.933	0.936	0.928	0.931†	0.936
Ionosphere	0.843	0.883	0.855	0.866	0.789	0.849	0.852	0.883	0.843	0.860	0.869	0.883	0.866	0.886	0.877	0.866	0.880	**0.886**	0.883
Iris	0.900	**0.967**	**0.967**	**0.967**	**0.967**	**0.967**	0.920	0.807	0.867	0.940	0.960	0.847	0.940	**0.967**	0.807	0.880	0.807	**0.967**	**0.967**
Liver-disorders	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**	**0.580**
Lung-cancer	0.510	0.529	0.562	0.529	0.467	**0.590**	0.471	0.538	0.471	0.438	0.500	0.505	0.505	0.381	0.562	0.500	0.533	0.510	0.500
Lymphography	0.845	0.783	0.757	0.784	0.831	0.838	0.764	0.858	0.790	0.804	0.804	**0.865**	0.689	0.791	0.824	0.858	0.844	0.804	0.838
Promoters	0.887	0.623	0.813	**0.887**	0.766	0.868	0.688	0.735	0.802	0.849	0.877	0.718	0.481	0.813	0.831	0.839	0.803	0.728	0.877
Monks-1-train	0.733	0.733	0.660	0.710	0.734	0.741	**0.743**	0.693	0.652	0.668	0.710	0.733	0.734	0.678	0.693	0.742	0.733	0.671	0.733
Monks-2-train	0.610	0.598	**0.621**	**0.621**	0.598	**0.621**	0.610	0.598	0.610	**0.621**	**0.621**	0.610	0.598	**0.621**	**0.621**	0.598	0.598	0.610	0.604
Monks-3-train	**0.934**	**0.934**	0.763	0.762	0.731	0.787	0.828	0.797	0.613	0.844	0.778	0.878	0.731	0.771	0.665	0.909	**0.934**	0.747	**0.934**
Parkinsons	0.835	0.768	0.850	0.763	0.850	0.825	**0.871**	0.861	0.866	0.855	**0.871**	0.851	0.851	0.804	0.861	0.861	0.861	0.784	**0.871**
Pima	0.737	0.671	0.669	0.673	0.678	0.655	0.680	0.674	0.746	0.663	0.758	0.758	0.763	0.655	0.681	0.704	0.684	0.751	**0.768**
Post-Operative	**0.711**	**0.711**	**0.711**	**0.711**	**0.711**	**0.711**	0.700	0.689	**0.711**	**0.711**	0.689	0.700	**0.711**	**0.711**	**0.711**	**0.711**	**0.711**	0.678	0.689
Segment	0.870	0.784	0.804	0.795	0.808	0.919	0.831	0.870	0.921	**0.929**	0.804	0.925	0.592	0.819	0.729	0.821	0.721	0.820†	0.929
Sonar	0.788	0.740	0.793	0.798	0.755	0.755	0.760	0.784	0.793	0.798	0.774	0.774	0.746	0.798	**0.803**	0.793	0.783	0.765	0.755
Soybean	0.934	0.884	0.886	0.871	0.892	**0.944**	0.898	0.884	0.890	0.936	0.873	0.870	0.862	0.898	0.889	0.896	0.892	0.896	0.928
Sponge	0.961	0.961	0.935	0.961	**0.973**	0.948	0.948	0.961	0.948	0.961	0.948	**0.973**	0.961	0.935	0.948	0.961	0.948	0.948	0.961
Tae	0.377	0.337	0.530	0.437	0.503	0.523	0.464	0.436	0.516	0.491	0.543	0.503	0.370	0.451	0.530	0.370	0.490	0.450	**0.549**
Thoracic	0.849	**0.851**	**0.851**	**0.851**	**0.851**	0.849	**0.851**	**0.851**	**0.851**	**0.851**	**0.851**	**0.851**	0.849	**0.851**	**0.851**	0.849	**0.851**	**0.851**	0.849
Tic-tac-toe	0.653	0.653	0.751	0.653	0.693	0.717	0.705	0.682	0.719	0.728	0.653	0.695	0.653	0.711	0.653	0.653	0.653	0.699	**0.983**
Silhouettes	0.668	0.664	0.632	0.676	0.667	0.687	0.609	0.669	0.689	0.684	0.595	0.708	0.577	0.668	0.675	0.673	0.675	0.647	**0.725**
Vote	**0.956**	**0.956**	**0.956**	**0.956**	0.899	**0.956**	0.908	0.880	**0.956**	**0.956**	**0.956**	**0.956**	0.736	**0.956**	**0.956**	**0.956**	**0.956**	0.949	**0.956**
Vowel	0.513	0.569	0.561	0.355	0.366	0.316	0.283	0.517	0.317	0.449	0.559	0.574	0.569	0.337	0.288	0.522	0.384	0.431	**0.676**
Wdbc	0.956	0.945	0.972	0.944	0.965	0.947	0.958	0.965	0.965	0.965	0.951	0.967	0.963	0.956	0.965	0.967	0.965	0.956	**0.977**
Wine	0.955	0.932	0.955	0.955	0.955	0.966	0.933	0.927	0.966	0.972	0.926	0.966	0.921	0.966	0.944	0.961	0.944	0.938	**0.989**
Zoo	0.901	0.940	0.941	0.941	0.891	0.911	0.871	0.841	0.911	0.940	0.921	0.911	0.831	0.940	0.881	0.911	0.881	0.920	**0.960**
Splice	0.941	0.597	0.946	0.948	0.936	0.942	0.704	0.711	0.909	0.936	0.879	0.915	0.525	**0.949**	0.946	0.946	0.946	0.683	0.930
Spambase	0.843	0.796	0.857	0.870	0.855	0.880	0.882	0.834	0.871	0.858	0.851	0.872	0.889	0.849	0.831	0.851	0.833	0.854†	**0.904**
Mushroom	1.000	**1.000**	**1.000**	**1.000**	0.994	**1.000**	0.995	0.999	0.995	1.000	**1.000**	0.998	0.891	0.984	0.999	1.000	1.000	0.991†	**1.000**
Optdigits	0.962	0.870	0.967	0.973	0.941	**1.000**	0.971	0.931	0.977	0.977	0.958	0.974	0.979	0.963	0.927	0.944	0.926	0.955†	0.982
Waveform-5000	0.816	0.870	0.870	0.871	0.538	**0.979**	0.663	0.603	0.857	0.870	0.867	0.870	0.808	0.871	0.796	0.816	0.792	0.809†	0.862
Pendigits	0.930	0.905	0.917	0.922	0.924	0.888	0.848	0.954	0.938	0.930	0.934	0.898	0.942	0.924	0.941	0.930	0.940	0.922†	**0.979**
Clean2	0.933	0.889	0.933	0.909	0.939	0.921	0.908	0.932	0.939	0.938	0.931	0.935	0.936	0.931	0.936	0.933	0.935	0.928†	**0.949**
Nursery	0.557	0.605†	0.684	0.688	0.710	0.605†	0.585	0.511	0.749	0.674	0.687	0.492	0.492	0.605†	0.634	0.501	0.504	0.605†	**0.932**
Average Rank	**9.35**	**11.31**	**8.78**	**8.23**	**10.46**	**7.87**	**10.82**	**11.35**	**8.7**	**7.48**	**9.8**	**6.77**	**11.64**	**9.42**	**9.79**	**9.1**	**9.58**	**10.55**	–

Table 11
Classification accuracy results of filter UFS methods on the UCI datasets of Table 1 using kNN ($k=3$).

Dataset	*Univariate*						*Multivariate*												
	Variance	SUD	LS	SVD-Entropy	SPEC	USFSM	FSFS	MC	MCFS	MRSF	UDFS	NDFS	RRFS	JELSR	UFSACO	MGSACO	RRFSACO	DSRMR	**Original**
Inflammations	0.883	**1.000**	**1.000**	0.975	**1.000**	**1.000**	0.967	0.883	0.925	0.958	0.925	**1.000**	0.650	0.958	0.900	0.900	0.942	0.900	**1.000**
Audiology	0.757	0.367	0.655	0.774	0.518	0.752	0.757	0.385	0.726	0.672	0.380	0.757	0.752	0.646	**0.792**	0.752	0.774	0.345	0.757
Automovile	0.741	0.737	0.698	0.668	0.683	0.741	**0.751**	0.727	0.722	0.737	0.663	0.727	0.741	0.673	0.737	0.722	0.727	0.727	**0.751**
Breast-cancer	0.703	0.696	0.734	0.752	0.710	0.713	0.720	0.685	0.738	**0.755**	0.731	0.689	0.692	0.741	0.737	0.699	0.738	0.710	0.731
Contraception	0.467	0.485	0.424	**0.490**	0.431	0.450	0.462	0.483	0.456	0.445	0.431	0.475	0.472	0.421	0.480	0.467	0.476	0.430	0.444
Credit-approval	0.667	0.816	**0.868**	0.854	0.816	0.852	0.835	0.701	0.841	0.838	0.764	0.845	0.765	0.859	0.675	0.706	0.671	0.725	0.816
Credit-german	0.710	0.680	0.660	0.635	0.689	0.672	0.702	0.683	0.649	0.701	0.687	0.690	0.645	0.702	0.655	0.706	0.670	0.675	**0.716**
Cylinder-bands	0.778	0.713	0.769	0.772	0.719	0.765	**0.783**	0.635	0.757	0.774	0.711	0.763	0.706	0.778	0.689	0.770	0.709	0.720	0.780
Dermatology	0.893	0.858	0.882	0.869	0.850	0.904	0.820	0.874	0.888	0.934	0.877	0.951	0.893	**0.959**	0.880	0.915	0.885	0.924	0.937
Ecoli	0.646	0.694	0.759	0.744	0.673	0.697	0.685	0.759	0.798	0.709	0.691	0.741	0.708	0.714	0.598	0.714	**0.816**	0.694	**0.816**
Flags	**0.567**	0.356	0.397	0.557	0.423	0.500	0.458	0.448	0.402	0.515	0.402	0.541	0.444	0.489	0.412	0.536	0.417	0.505	0.500
Heart-c	0.756	0.772	0.795	0.766	0.749	**0.805**	0.763	0.706	0.756	0.746	0.776	0.719	0.743	0.766	0.759	0.726	0.749	0.723	0.759
Heart-h	0.776	0.833	0.840	0.789	0.834	0.796	0.755	0.677	0.810	0.806	0.840	0.793	0.755	**0.844**	0.728	0.745	0.731	0.728	0.765
Heart-statlog	0.759	0.700	0.789	0.781	0.774	**0.796**	0.741	0.711	0.730	0.722	0.763	0.789	0.785	0.733	0.719	0.752	0.737	0.763	0.759
Hepatitis	0.826	0.787	0.832	0.819	0.787	0.787	0.794	0.781	**0.845**	0.813	0.800	0.832	0.794	0.806	0.781	0.826	0.813	0.839	0.800
Horse-colic	0.677	0.690	0.775	0.653	0.690	0.728	0.788	0.774	0.804	0.755	0.777	**0.815**	0.739	0.777	0.685	0.685	0.706	0.807	0.747
Hypothyroid	0.918	0.954	0.923	0.922	0.927	0.917	0.902	**0.969**	0.923	0.932	0.920	0.933	0.928	0.923	0.955	0.923	0.940	0.930†	0.914
Ionosphere	0.880	0.877	0.883	0.889	0.877	0.874	0.883	0.895	0.897	0.892	0.895	0.900	**0.906**	0.900	0.895	0.877	0.897	0.897	0.874
Iris	0.900	**0.967**	**0.967**	**0.967**	**0.967**	**0.967**	0.900	0.707	0.813	0.933	**0.967**	0.787	0.933	**0.967**	0.707	0.853	0.707	**0.967**	0.947
Liver-disorders	0.629	0.643	0.609	0.641	0.641	0.641	0.594	0.603	0.603	0.591	0.617	**0.652**	0.609	0.612	0.571	0.626	0.583	0.641	0.609
Lung-cancer	0.471	0.533	0.438	0.471	0.562	0.467	0.419	0.567	0.533	0.376	0.471	0.500	0.443	0.433	**0.590**	0.562	0.471	0.433	0.500
Lymphography	0.777	0.756	0.777	0.757	0.804	0.819	0.697	**0.831**	0.764	0.803	0.735	0.817	0.676	0.797	0.750	0.763	0.743	0.737	0.804
Promoters	**0.812**	0.680	0.736	0.736	0.719	0.736	0.632	0.736	0.792	0.736	0.774	0.766	0.481	0.792	0.755	0.690	0.729	0.727	0.794
Monks-1-train	**0.911**	**0.911**	0.579	0.789	0.678	0.694	0.711	0.702	0.621	0.603	0.670	0.886	0.734	0.622	0.734	0.869	**0.911**	0.655	0.758
Monks-2-train	0.598	0.598	0.604	0.585	0.610	0.597	0.627	**0.639**	0.633	0.622	0.639	0.592	0.598	0.586	0.586	0.598	0.568	0.622	0.609
Monks-3-train	**0.918**	**0.918**	0.656	0.721	0.698	0.745	0.761	0.740	0.606	0.680	0.655	0.845	0.731	0.698	0.583	0.868	0.910	0.713	0.836
Parkinsons	0.902	0.887	0.948	0.882	0.953	0.923	0.958	0.953	0.964	0.943	0.959	0.928	0.923	0.897	0.953	0.954	0.948	0.871	**0.969**
Pima	0.665	0.647	0.624	0.641	0.648	0.639	0.646	0.609	0.676	0.625	0.672	0.678	0.672	0.633	0.658	0.652	0.639	0.677	**0.701**
Post-Operative	**0.711**	0.678	**0.711**	0.700	**0.711**	**0.711**	0.667	0.689	**0.711**	**0.711**	0.689	0.700	**0.711**	**0.711**	**0.711**	**0.711**	**0.711**	0.689	0.600
Segment	0.953	0.915	0.919	0.917	0.919	**0.972**	0.943	0.955	0.965	0.966	0.919	0.968	0.751	0.921	0.892	0.938	0.893	0.924†	0.966
Sonar	0.870	0.856	0.875	0.856	0.798	0.846	0.826	**0.880**	0.870	0.861	0.817	0.856	0.861	0.875	0.827	0.856	0.861	0.861	0.856
Soybean	0.870	0.883	0.886	0.868	0.887	0.905	0.849	0.843	0.893	0.917	0.868	0.873	0.830	0.899	0.877	0.858	0.895	0.874	**0.921**
Sponge	**0.973**	**0.973**	**0.973**	0.948	0.960	0.948	0.921	0.922	0.934	0.921	0.948	**0.973**	0.921	0.935	0.935	0.961	0.948	0.948	0.921
Tae	0.563	0.577	0.529	0.437	0.556	0.509	0.544	0.603	0.503	0.471	0.523	0.556	0.563	0.470	0.569	0.563	0.589	0.549	**0.630**
Thoracic	0.770	0.751	0.768	0.764	0.804	0.838	0.779	0.800	0.806	**0.847**	0.774	0.823	0.766	0.845	0.768	0.762	0.774	0.796	0.764
Tic-tac-toe	0.637	0.647	0.802	0.658	0.762	0.730	0.742	0.734	0.752	0.739	0.660	0.686	0.653	0.737	0.667	0.635	0.660	0.718	**0.990**
Silhouettes	0.687	0.690	0.673	0.712	0.657	0.686	0.639	0.668	0.709	0.688	0.689	**0.720**	0.602	0.696	0.660	0.662	0.660	0.675	0.702
Vote	0.936	0.956	0.949	0.954	0.885	0.954	0.903	0.867	0.947	0.945	**0.959**	0.947	0.759	0.956	0.940	0.929	0.940	0.929	0.926
Vowel	0.976	0.985	0.971	0.951	0.965	0.945	0.879	0.973	0.951	0.966	0.969	0.969	0.981	0.941	0.940	0.980	0.956	0.970	**0.987**
Wdbc	0.945	0.942	**0.961**	0.933	0.960	0.947	0.947	0.946	0.960	0.958	0.942	0.960	0.947	0.946	0.951	0.951	0.951	0.947	0.958
Wine	0.955	0.899	0.955	0.944	0.955	0.949	0.905	0.888	0.967	0.972	0.944	**0.989**	0.944	0.966	0.922	0.961	0.938	0.915	0.933
Zoo	0.901	0.950	0.940	0.950	0.901	0.891	0.861	0.840	0.902	0.930	0.940	0.950	0.880	0.930	0.892	0.891	0.892	0.950	**0.960**
Splice	0.758	0.575	0.765	**0.773**	0.735	0.770	0.646	0.606	0.726	0.749	0.697	0.724	0.525	0.762	0.767	0.758	0.757	0.683	0.741
Spambase	0.890	0.889	0.891	0.881	**0.903**	0.890	0.895	0.889	0.884	0.877	0.879	0.893	0.894	0.887	0.879	0.885	0.879	0.887†	**0.903**
Mushroom	**1.000**	**1.000**	**1.000**	**1.000**	0.994	**1.000**	0.997	0.999	0.998	**1.000**	**1.000**	**1.000**	0.897	0.998	**1.000**	**1.000**	**1.000**	0.993†	**1.000**
Optdigits	0.974	0.849	0.975	0.980	0.947	0.985	0.980	0.937	0.983	0.985	0.967	0.979	0.986	0.968	0.941	0.947	0.938	0.960†	**0.986**
Waveform-5000	0.692	0.787	**0.798**	0.796	0.442	0.796	0.501	0.442	0.750	0.788	0.786	0.790	0.686	0.793	0.652	0.678	0.656	0.696†	0.730
Pendigits	0.976	0.964	0.970	0.978	0.963	0.972	0.929	0.986	0.975	0.973	0.977	0.955	0.982	0.971	0.980	0.976	0.975	0.971†	**0.993**
Clean2	0.944	0.957	0.942	0.952	0.948	0.961	0.941	0.936	0.957	0.958	0.954	**0.964**	0.946	0.953	0.948	0.949	0.948	0.950†	0.955
Nursery	0.556	0.615†	0.703	0.695	0.744	0.615†	0.592	0.507	0.771	0.691	0.706	0.484	0.492	0.615†	0.649	0.499	0.520	0.615†	**0.981**
Average Rank	**8.83**	**9.88**	**8.24**	**9.11**	**9.99**	**8.1**	**11.33**	**11.23**	**8.05**	**8.41**	**9.92**	**6.67**	**11.27**	**8.35**	**11.18**	**9.8**	**10.29**	**10.35**	–

Table 12
Classification accuracy results of filter UFS methods on the UCI datasets of Table 1 using NB.

Dataset	*Univariate*						*Multivariate*												
	Variance	SUD	LS	SVD-Entropy	SPEC	USFSM	FSFS	MC	MCFS	MRSF	UDFS	NDFS	RRFS	JELSR	UFSACO	MGSACO	RRFSACO	DSRMR	**Original**
Inflammations	0.833	0.992	0.992	0.983	**1.000**	0.917	0.858	0.833	0.867	0.900	0.933	**1.000**	0.833	0.925	0.867	0.850	0.867	0.825	**1.000**
Audiology	0.721	0.296	0.624	0.726	0.464	0.703	0.726	0.323	0.668	0.650	0.301	**0.730**	0.695	0.606	0.672	0.668	0.681	0.292	0.712
Automovile	0.639	0.610	0.541	0.556	0.483	0.634	0.654	0.668	0.620	0.644	0.551	0.615	0.668	0.551	0.693	0.654	**0.702**	0.595	0.673
Breast-cancer	0.717	0.738	0.738	0.731	0.731	0.742	0.731	0.685	**0.745**	0.741	0.713	0.713	0.692	0.717	0.738	0.727	0.713	0.703	0.734
Contraception	0.513	0.480	0.436	0.504	0.444	0.451	0.460	0.458	0.490	0.447	0.446	**0.514**	0.506	0.444	0.498	0.510	0.511	0.430	0.498
Credit-approval	0.793	0.862	0.861	0.849	0.862	0.862	0.857	0.790	**0.868**	0.848	0.813	0.854	0.843	0.861	0.783	0.799	0.778	0.787	0.864
Credit-german	**0.745**	0.742	0.707	0.714	0.698	0.723	0.743	0.708	0.703	0.727	0.707	0.703	0.699	0.721	0.721	0.738	0.712	0.740	0.739
Cylinder-bands	0.770	0.715	0.752	0.750	0.689	0.752	0.743	0.704	0.737	0.754	0.724	0.759	0.700	0.756	0.707	**0.781**	0.717	0.726	0.750
Dermatology	0.959	0.869	0.899	0.888	0.904	0.929	0.902	0.970	0.899	0.934	0.896	0.978	0.951	**0.984**	0.924	0.932	0.924	0.948	0.984
Ecoli	0.690	0.655	0.792	0.789	0.673	0.744	0.735	0.789	0.795	0.768	0.726	0.795	0.756	0.753	0.666	0.753	**0.828**	0.655	**0.828**
Flags	**0.691**	0.438	0.521	0.644	0.433	0.618	0.587	0.546	0.562	0.598	0.454	0.608	0.531	0.618	0.525	0.664	0.495	0.644	0.680
Heart-c	0.795	0.835	0.838	0.805	0.825	**0.842**	0.815	0.792	0.762	0.799	0.839	0.815	0.805	0.785	0.808	0.809	0.805	0.802	0.842
Heart-h	0.830	0.834	**0.844**	0.837	0.837	0.834	0.806	0.714	0.840	0.834	0.837	0.834	0.823	0.837	0.769	0.810	0.759	0.827	0.840
Heart-statlog	0.807	0.774	0.833	0.811	0.811	0.833	0.815	0.833	0.811	0.793	0.800	0.811	**0.837**	0.815	0.815	0.781	0.815	0.815	0.807
Hepatitis	0.806	0.845	0.845	0.852	0.826	0.839	0.839	0.839	**0.871**	0.839	0.845	0.845	0.800	0.852	0.852	0.806	0.826	0.845	0.839
Horse-colic	0.761	0.726	0.818	0.742	0.756	0.810	0.810	0.832	0.824	0.791	0.818	**0.840**	0.783	0.829	0.758	0.769	0.788	0.823	0.805
Hypothyroid	0.983	0.967	0.923	0.982	0.974	0.962	0.932	0.979	0.923	0.933	0.974	0.985	0.985	0.923	0.975	**0.986**	0.956	0.961†	0.984
Ionosphere	0.895	0.900	0.897	0.892	0.875	0.875	0.912	0.906	0.889	0.900	0.892	0.889	0.892	0.889	0.900	0.889	0.900	**0.914**	0.895
Iris	0.907	**0.953**	**0.953**	**0.953**	**0.953**	**0.953**	0.900	0.693	0.840	0.947	**0.953**	0.793	0.947	**0.953**	0.693	0.860	0.693	**0.953**	0.940
Liver-disorders	0.574	0.565	**0.580**	**0.580**	0.565	0.565	**0.580**	**0.580**	**0.580**	**0.580**	0.565	0.565	**0.580**	0.574	**0.580**	0.574	**0.580**	0.565	0.565
Lung-cancer	0.538	**0.624**	0.571	0.586	0.462	0.562	0.443	0.557	0.571	0.595	0.533	0.505	0.567	0.495	**0.624**	0.505	0.533	0.505	0.529
Lymphography	0.831	0.769	0.722	0.750	0.804	0.824	0.723	0.832	0.797	0.797	0.784	0.824	0.649	0.764	0.762	0.764	0.776	0.797	**0.838**
Promoters	**0.897**	0.689	0.869	0.870	0.728	0.887	0.718	0.765	0.859	0.887	0.859	0.812	0.481	0.860	0.841	0.832	0.841	0.784	0.868
Monks-1-train	0.726	0.742	0.635	0.694	0.710	0.749	0.677	0.654	0.636	0.643	0.702	0.733	0.734	0.678	0.670	0.726	**0.758**	0.629	0.750
Monks-2-train	**0.633**	0.616	0.621	0.621	0.627	0.616	0.622	0.622	0.621	0.621	0.621	0.622	0.598	0.621	0.627	0.627	0.604	0.598	0.574
Monks-3-train	**0.934**	**0.934**	0.755	0.770	0.731	0.787	0.828	0.789	0.588	0.836	0.762	0.878	0.731	0.780	0.656	0.901	**0.934**	0.739	**0.934**
Parkinsons	0.819	0.794	0.758	0.789	0.820	0.748	0.825	0.830	0.825	0.820	0.820	0.809	**0.856**	0.804	0.825	0.830	0.830	0.799	0.794
Pima	0.727	0.697	0.677	0.691	0.693	0.691	0.693	0.669	**0.750**	0.674	0.715	0.736	0.747	0.677	0.689	0.699	0.691	0.747	0.741
Post-Operative	0.711	0.711	0.711	0.711	**0.722**	0.711	0.700	0.700	0.700	**0.722**	0.678	0.700	0.711	0.711	0.711	0.711	0.711	0.689	0.667
Segment	0.887	0.796	0.823	0.832	0.848	0.887	0.874	0.900	0.919	0.926	0.822	**0.930**	0.692	0.823	0.826	0.868	0.829	0.852†	0.912
Sonar	0.774	0.755	0.774	0.760	0.701	0.716	0.726	0.774	0.765	**0.803**	0.755	0.769	0.765	0.774	0.784	0.798	0.764	0.740	0.765
Soybean	0.877	0.858	0.840	0.848	0.842	0.899	0.854	0.840	0.861	0.912	0.857	0.845	0.808	0.840	0.839	0.874	0.862	0.854	**0.915**
Sponge	**0.987**	0.868	0.883	0.855	0.948	0.882	0.922	0.948	0.935	0.897	0.922	0.948	0.934	0.948	0.935	0.948	0.934	0.948	0.935
Tae	0.344	0.344	**0.470**	0.457	0.411	**0.470**	0.437	0.364	0.437	**0.470**	**0.470**	0.411	0.344	**0.470**	0.431	0.344	0.398	0.378	**0.470**
Thoracic	0.847	0.851	0.851	0.851	0.851	0.836	0.847	0.851	0.845	0.849	0.851	0.851	0.847	**0.853**	0.851	0.851	**0.853**	0.849	0.826
Tic-tac-toe	0.658	0.659	0.721	0.649	0.707	0.700	0.698	0.704	0.719	**0.731**	0.666	0.697	0.653	0.707	0.655	0.662	0.656	0.713	0.706
Silhouettes	0.556	0.586	0.563	0.559	0.556	0.558	0.587	**0.632**	0.602	0.606	0.567	0.604	0.539	0.576	0.610	0.580	0.610	0.600	0.598
Vote	0.899	0.920	0.940	0.920	0.871	0.917	0.897	0.867	**0.945**	**0.945**	0.917	0.924	0.756	0.922	0.938	0.910	0.936	0.922	0.901
Vowel	0.439	0.552	0.510	0.378	0.314	0.330	0.352	0.492	0.326	0.400	0.515	0.541	0.535	0.330	0.300	0.475	0.360	0.371	**0.574**
Wdbc	0.931	0.942	0.945	0.944	0.939	0.947	0.935	0.949	0.951	0.945	0.938	0.937	0.932	0.947	0.940	0.949	**0.953**	0.937	0.940
Wine	0.949	0.938	0.977	0.972	0.944	0.983	0.966	0.938	0.972	0.972	0.915	0.983	0.944	0.972	**0.983**	0.955	**0.983**	0.949	0.983
Zoo	0.901	**0.950**	0.950	0.940	0.911	0.900	0.880	0.841	0.911	0.930	0.921	0.920	0.871	0.940	0.871	0.901	0.871	0.920	0.950
Splice	0.956	0.594	**0.961**	0.958	0.948	0.951	0.712	0.703	0.912	0.950	0.889	0.934	0.525	0.959	0.956	0.955	0.956	0.820	0.954
Spambase	0.892	0.892	0.894	0.897	**0.909**	0.888	0.876	0.880	0.875	0.870	0.884	0.895	0.889	0.885	0.870	0.881	0.870	0.885†	0.899
Mushroom	0.955	0.963	**0.989**	0.920	0.930	0.952	0.982	0.985	0.950	0.976	**0.989**	0.920	0.889	0.897	0.919	0.966	0.975	0.951†	0.954
Optdigits	0.895	0.792	0.894	0.902	0.864	0.911	0.905	0.865	0.915	0.910	0.890	0.903	0.915	0.878	0.867	0.874	0.863	0.885†	**0.921**
Waveform-5000	0.774	0.802	0.804	0.803	0.535	0.803	0.644	0.594	0.798	**0.804**	0.804	0.802	0.781	0.803	0.768	0.782	0.768	0.757†	0.802
Pendigits	0.832	0.805	0.829	0.818	0.818	0.826	0.810	0.873	0.860	0.856	0.866	0.811	0.859	0.841	0.851	0.839	0.854	0.838†	**0.877**
Clean2	0.901	0.901	0.905	0.900	0.908	0.899	0.891	0.885	0.898	0.902	0.909	**0.909**	0.893	0.907	0.901	0.904	0.902	0.901†	0.908
Nursery	0.564	0.613†	0.694	0.699	0.725	0.613†	0.593	0.518	0.762	0.683	0.699	0.494	0.492	0.613†	0.642	0.504	0.520	0.613†	**0.902**
Average Rank	**8.88**	**10.32**	**8.3**	**9.06**	**11.13**	**8.64**	**10.62**	**10.24**	**8.65**	**7.58**	**9.96**	**7.83**	**11.52**	**8.75**	**10.07**	**8.96**	**9.4**	**11.09**	–

Table 13
Classification accuracy results of filter UFS methods on the ASU datasets of Table 2 using SVM.

Dataset	*Univariate*						*Multivariate*												
	Variance	SUD	LS	SVD-Entropy	SPEC	USFSM	FSFS	MC	MCFS	MRSF	UDFS	NDFS	RRFS	JELSR	UFSACO	MGSACO	RRFSACO	DSRMR	**Original**
Yale	0.733	0.709	0.727	0.709	0.691	0.745	0.606	0.727	0.752	0.709	0.752	0.739	0.745	0.733	0.745	**0.794**	0.745	0.764	0.764
Isolet	0.955	0.957	0.958	0.958	**0.963**	0.958	0.938	0.954	0.957	**0.963**	0.960	0.960	0.959	0.958	0.960	0.959	0.960	0.957	0.962
ORL	0.970	0.953	0.963	0.953	0.950	**0.975**	0.880	0.960	0.973	0.970	0.970	0.965	0.963	0.970	0.968	0.973	0.968	0.970	0.973
WarpPIE10P	0.990	0.995	0.986	0.995	0.919	**1.000**	0.819	0.995	0.990	0.995	0.990	0.990	**1.000**	0.990	0.986	0.995	0.986	0.995	**1.000**
Colon	0.774	0.756	0.742	0.659	0.822	0.773	0.742	0.774	0.822	0.724	0.822	0.790	0.805	**0.837**	0.758	0.726	0.773	0.724	0.788
WarpAR10P	0.931	0.700	0.700	0.792	0.900	0.985	0.446	0.969	0.977	0.908	0.992	0.977	0.962	0.977	0.992	0.992	0.992	0.985	**1.000**
Lung	0.936	0.961	0.956	0.951	0.966	0.956	0.931	0.946	**0.966**	0.941	0.951	**0.966**	0.961	0.946	0.961	0.966	0.961	0.956	**0.966**
COIL20	0.999	0.990	0.981	0.994	0.999	**1.000**	0.938	0.997	0.999	0.999	0.995	0.999	0.999	0.996	0.998	0.999	0.999	0.998	**1.000**
Lymphoma	0.937	0.906	0.927	0.906	0.916	0.937	0.594	0.917	0.947	0.917	0.937	0.937	**0.958**	0.875	0.937	**0.958**	0.937	0.916	**0.958**
GLIOMA	0.720	0.780	0.820	0.800	0.780	0.780	0.640	0.740	0.800	0.620	0.780	0.800	0.780	0.720	**0.860**	0.840	0.820	0.780	0.780
ALLAML	0.848	0.945	0.905	0.890	0.875	0.917	0.790	0.889	0.931	0.764	0.890	0.903	0.959	0.930	0.958	0.931	0.946	0.946	**0.987**
CLL_SUB_111	0.541	0.676†	0.667	0.676†	0.712	0.703	0.585	0.675	0.630	0.659	0.667	0.702	0.684	0.667	0.730	0.738	0.729	0.729	**0.757**
Carcinom	0.758	0.870†	0.919	0.870†	0.896	0.885	0.770	0.880	**0.925**	0.827	0.868	0.897	0.885	0.845	0.891	0.903	0.896	0.873	**0.925**
GLI_85	0.812	0.838†	0.871	0.838†	0.894	0.871	0.682	0.776	0.882	0.741	**0.918**	0.800	0.835	0.838	0.859	0.882	0.906	0.838	0.906
Prostate_GE	0.864	0.863†	0.853	0.863†	0.902	0.892	0.883	0.843	0.882	0.765	0.854	0.813	0.873	0.794	0.922	0.893	**0.941**	0.824	0.892
SMK_CAN_187	0.690	0.672†	0.684	0.672†	0.694	0.662	0.621	0.684	0.647	0.663	0.706	0.651	0.647	0.700	0.716	0.625	0.652	0.700	**0.722**
TOX_171	0.906	0.850†	0.900	0.850†	0.865	0.837	0.585	0.859	0.889	0.848	0.824	0.860	0.883	0.783	0.906	0.894	0.877	0.883	**0.959**
Arcene	0.575	0.703†	0.800	0.703†	0.820	0.475	0.745	0.785	0.770	0.585	0.640	0.650	0.785	0.660	0.740	0.730	0.730	0.760	**0.850**
Leukemia	0.914	0.907†	0.890	0.907†	0.916	0.959	0.849	0.861	0.958	0.877	0.917	0.918	0.973	0.848	0.890	0.890	0.902	0.947	**0.987**
Nci9	0.217	0.469†	0.433	0.469†	0.550	0.517	0.350	**0.583**	**0.583**	0.267	0.550	0.517	0.533	0.283	0.550	0.533	0.567	0.467	**0.583**
Orlraws10P	0.930	0.946†	0.950	0.946†	0.900	0.960	0.710	0.950	0.990	0.970	0.970	0.990	0.970	0.900	0.990	**1.000**	0.980	0.980	0.980
Pixraw10P	0.880	0.956†	0.950	0.956†	0.960	**0.990**	0.720	0.980	**0.990**	0.980	0.980	0.980	**0.990**	0.960	0.980	**0.990**	**0.990**	0.970	**0.990**
RELATHE	0.547	0.748†	0.776	0.748†	0.790	0.798	0.810	0.744	0.773	0.700	0.746	0.730	0.806	0.734	0.795	0.809	0.788	0.624	**0.885**
PCMAC	0.542	0.739†	0.701	0.739†	0.712	0.770	0.798	0.740	0.738	0.707	0.709	0.726	0.803	0.725	0.811	0.788	0.806	0.746	**0.850**
BASEHOCK	0.499	0.802†	0.773	0.802†	0.802	0.802	0.899	0.786	0.774	0.802†	0.802†	0.808	0.908	0.741	0.912	0.901	0.898	0.728	**0.946**
Average rank	**12.76**	**11.48**	**10.90**	**11.84**	**8.74**	**7.46**	**14.78**	**11.06**	**6.56**	**12.92**	**8.92**	**8.78**	**6.38**	**12.40**	**5.64**	**5.18**	**6.08**	**9.12**	–

Table 14
Classification accuracy results of filter UFS methods on the ASU datasets of Table 2 using kNN ($k = 3$).

Dataset	*Univariate*						*Multivariate*												
	Variance	SUD	LS	SVD-Entropy	SPEC	USFSM	FSFS	MC	MCFS	MRSF	UDFS	NDFS	RRFS	JELSR	UFSACO	MGSACO	RRFSACO	DSRMR	**Original**
Yale	0.618	0.648	0.642	0.648	0.606	0.636	0.479	0.630	**0.673**	0.642	0.642	0.655	0.655	0.630	0.655	**0.673**	0.648	0.606	**0.673**
Isolet	0.874	0.880	0.882	0.874	**0.896**	0.881	0.826	0.872	0.878	0.892	0.860	0.894	0.871	0.865	0.878	0.879	0.883	0.869	0.886
ORL	0.925	0.918	0.930	0.920	0.918	0.930	0.840	0.943	0.933	0.943	0.933	0.930	0.938	0.925	0.930	0.935	0.928	0.940	**0.948**
WarpPIE10P	0.967	0.933	0.905	0.919	0.971	0.990	0.919	**0.995**	0.986	0.990	**0.995**	0.971	0.981	**0.995**	**0.995**	0.990	**0.995**	0.938	0.990
Colon	0.681	0.594	0.628	0.613	0.710	0.709	**0.713**	0.696	0.664	0.644	0.694	0.710	0.664	0.646	0.695	0.696	0.696	0.631	0.663
WarpAR10P	0.538	0.454	0.415	0.577	**0.808**	0.469	0.500	0.692	0.477	0.415	0.523	0.462	0.562	0.592	0.508	0.485	0.500	0.408	0.492
Lung	0.912	0.941	0.951	**0.956**	0.946	0.912	0.907	0.897	0.936	0.907	0.911	0.941	0.936	0.897	0.941	0.936	0.931	0.931	0.936
COIL20	0.997	0.985	0.974	0.985	0.996	**0.999**	0.988	**0.999**	0.998	**0.999**	0.987	**0.999**	0.999	0.990	**0.999**	**0.999**	**0.999**	0.995	**0.999**
Lymphoma	0.906	0.885	0.906	0.885	0.885	**0.968**	0.542	0.844	0.937	0.864	0.948	0.927	0.948	0.864	0.947	0.948	0.927	0.906	0.948
GLIOMA	0.460	0.760	0.700	0.760	0.760	0.700	0.640	0.680	**0.780**	0.500	0.740	0.740	0.680	0.560	0.680	0.660	0.680	**0.780**	0.700
ALLAML	0.791	0.874	0.862	0.762	0.848	0.831	0.764	0.791	**0.891**	0.639	0.759	0.836	0.806	0.750	0.889	0.875	0.846	0.790	0.874
CLL_SUB_111	0.468	0.541†	0.540	0.541†	0.549	0.586	0.558	0.515	0.566	0.577	0.514	0.523	0.585	0.477	0.568	0.549	0.541	0.540	**0.621**
Carcinom	0.557	0.780†	0.850	0.780†	0.833	0.793	0.747	0.799	**0.902**	0.753	0.758	0.793	0.828	0.615	0.810	0.856	0.810	0.776	0.862
GLI_85	0.659	0.794†	0.788	0.794†	0.871	0.847	0.776	0.576	0.871	0.718	0.812	0.859	0.871	0.706	0.812	0.824	**0.882**	0.835	**0.882**
Prostate_GE	0.695	0.768†	0.717	0.768†	**0.831**	0.793	0.794	0.801	0.792	0.746	0.743	0.726	0.792	0.745	0.773	0.775	0.793	0.765	0.823
SMK_CAN_187	0.626	0.628†	0.647	0.628†	0.609	0.637	0.572	0.588	0.604	**0.690**	0.657	0.619	0.642	0.610	0.631	0.657	0.631	0.625	0.647
TOX_171	0.748	0.792†	0.795	0.792†	0.830	0.719	0.691	0.866	**0.918**	0.813	0.830	0.830	0.701	0.696	0.854	0.818	0.865	0.702	0.836
Arcene	0.550	0.708†	0.820	0.708†	0.800	0.470	0.765	0.785	0.815	0.630	0.625	0.720	0.770	0.660	0.710	0.725	0.705	0.785	**0.865**
Leukemia	0.847	0.816†	0.846	0.816†	0.849	0.862	0.695	0.764	0.917	0.778	0.889	0.847	**0.930**	0.668	0.791	0.780	0.790	0.806	0.917
Nci9	0.133	0.403†	0.333	0.403†	0.467	0.350	0.467	**0.567**	0.467	0.233	0.450	0.467	0.533	0.150	**0.567**	0.500	0.467	0.300	0.450
Orlraws10P	0.830	0.908†	0.960	0.908†	0.910	0.920	0.730	0.910	0.960	0.890	0.900	0.930	0.960	0.810	0.950	0.940	0.950	**0.980**	0.930
Pixraw10P	0.850	0.969†	0.980	0.969†	**0.990**	**0.990**	0.920	0.970	**0.990**	0.980	0.980	**0.990**	0.980	0.930	**0.990**	**0.990**	**0.990**	0.980	**0.990**
RELATHE	0.567	0.726†	0.768	0.726†	0.746	0.750	**0.781**	0.733	0.732	0.657	0.698	0.721	0.777	0.755	0.760	0.741	0.752	0.683	0.772
PCMAC	0.629	0.681†	0.702	0.681†	0.667	0.687	0.666	**0.719**	0.679	0.665	0.675	0.655	0.680	0.704	0.707	0.697	0.690	0.676	0.712
BASEHOCK	0.502	0.739†	0.759	0.739†	0.739	0.739	0.768	0.764	0.735	0.739†	0.739†	0.731	0.777	0.725	0.787	**0.792**	0.779	0.751	0.776
Average rank	**14.22**	**11.10**	**9.12**	**11.08**	**7.42**	**8.06**	**12.78**	**9.02**	**6.52**	**11.76**	**10.50**	**8.80**	**6.48**	**13.72**	**6.06**	**6.36**	**6.90**	**11.10**	–

Table 15
Classification accuracy results of filter UFS methods on the ASU datasets of Table 2 using NB.

Dataset	*Univariate*						*Multivariate*												
	Variance	SUD	LS	SVD-Entropy	SPEC	USFSM	FSFS	MC	MCFS	MRSF	UDFS	NDFS	RRFS	JELSR	UFSACO	MGSACO	RRFSACO	DSRMR	**Original**
Yale	0.594	0.594	0.667	0.661	0.618	0.655	0.200	0.624	0.636	0.570	0.667	0.667	0.636	0.515	0.685	0.655	0.667	0.618	**0.709**
Isolet	0.888	0.876	0.883	0.883	**0.925**	0.892	0.867	0.877	0.884	0.903	0.881	0.899	0.883	0.879	0.878	0.883	0.879	0.888	0.896
ORL	0.830	0.790	0.813	0.790	0.708	0.870	0.515	0.865	0.888	**0.900**	0.890	**0.900**	0.858	0.880	0.863	0.880	0.868	0.818	0.880
WarpPIE10P	0.838	0.390	0.433	0.314	0.786	0.910	0.719	**0.986**	0.929	0.905	0.967	0.938	0.852	0.962	0.943	0.929	0.933	0.605	0.929
Colon	0.662	0.615	0.464	0.499	0.676	0.773	0.705	0.760	0.758	0.742	0.677	0.694	0.790	**0.806**	0.776	**0.806**	0.776	0.560	0.727
WarpAR10P	0.600	0.254	0.315	0.315	0.577	0.669	0.246	0.746	0.762	0.508	0.731	0.638	0.662	0.731	0.754	**0.769**	0.762	0.654	**0.769**
Lung	0.941	0.921	0.926	0.916	0.946	0.936	0.941	0.941	**0.970**	0.921	0.956	0.956	0.961	0.941	0.961	0.946	0.961	0.921	0.956
COIL20	0.970	0.923	0.893	0.936	0.969	0.983	0.934	0.985	0.980	0.984	0.927	0.981	0.985	0.962	**0.988**	0.985	0.987	0.953	0.985
Lymphoma	0.824	0.802	0.834	0.813	0.834	0.907	0.501	0.468	0.906	0.699	0.803	0.843	0.928	0.459	0.917	0.917	0.896	0.865	**0.938**
GLIOMA	0.720	0.720	0.740	0.720	0.760	0.800	0.580	0.760	0.780	0.540	0.760	0.740	0.800	0.640	0.760	0.760	0.800	0.740	**0.840**
ALLAML	0.903	0.889	0.833	0.750	0.902	0.917	0.820	0.930	0.932	0.721	0.846	0.902	0.903	0.946	**0.972**	0.945	0.945	0.807	0.958
CLL_SUB_111	0.623	0.648†	0.532	0.648†	0.551	0.693	0.629	**0.757**	0.683	0.649	0.632	0.694	0.729	0.614	0.658	0.685	0.667	0.568	0.676
Carcinom	0.742	0.789†	0.822	0.789†	0.839	0.816	0.655	0.748	0.908	0.689	0.782	0.868	0.856	0.592	0.851	0.885	0.839	0.729	**0.914**
GLI_85	0.812	0.820†	0.812	0.820†	0.835	0.765	0.788	0.812	0.847	0.765	**0.871**	0.788	0.859	0.812	0.835	0.835	0.859	0.824	0.859
Prostate_GE	0.814	0.799†	0.637	0.799†	0.794	**0.882**	0.833	0.831	0.873	0.697	0.813	0.745	0.873	0.766	0.854	0.854	0.873	0.646	0.833
SMK_CAN_187	0.604	0.663†	0.647	0.663†	0.674	0.685	0.593	0.668	0.695	0.680	0.679	0.690	0.668	0.663	0.674	0.663	**0.712**	0.620	0.641
TOX_171	0.636	0.616†	0.655	0.616†	0.620	0.690	0.439	0.479	0.625	0.596	0.666	0.585	0.637	0.572	0.666	0.643	**0.713**	0.637	0.701
Arcene	0.605	0.650†	**0.690**	0.650†	0.670	0.625	0.640	0.660	0.655	0.650	0.625	0.675	0.660	0.630	0.660	0.660	0.650	0.650	0.670
Leukemia	0.791	0.836†	0.833	0.836†	0.930	0.889	0.777	0.639	0.874	0.765	0.819	0.931	0.958	0.860	0.818	0.862	0.777	0.846	**0.959**
Nci9	0.167	0.211†	0.150	0.211†	0.267	0.183	0.217	0.233	0.250	0.133	0.200	0.267	0.383	0.167	0.250	0.217	0.183	0.117	**0.500**
Orlraws10P	0.880	0.939†	0.950	0.939†	0.960	0.910	0.730	0.960	**1.000**	0.930	0.970	0.990	0.970	0.910	0.970	0.980	0.970	0.950	0.990
Pixraw10P	0.960	0.977†	0.980	0.977†	0.970	0.990	0.930	0.980	0.990	**1.000**	0.980	**1.000**	0.980	0.940	0.990	0.990	0.990	0.960	0.980
RELATHE	0.560	0.684†	0.685	0.684†	0.684	0.717	0.760	0.546	0.694	0.631	0.633	0.648	0.760	0.633	0.780	0.774	0.769	0.670	**0.849**
PCMAC	0.629	0.718†	0.607	0.718†	0.621	0.796	0.832	0.505	0.693	0.747	0.695	0.680	0.854	0.607	0.823	0.785	0.833	0.783	**0.874**
BASEHOCK	0.501	0.773†	0.668	0.773†	0.773	0.773	0.921	0.553	0.744	0.773†	0.773†	0.765	0.929	0.623	0.945	0.927	0.941	0.754	**0.967**
Average rank	**13.10**	**12.76**	**12.28**	**12.26**	**9.62**	**7.40**	**13.74**	**10.20**	**5.80**	**11.84**	**9.06**	**7.22**	**5.24**	**12.38**	**5.12**	**5.32**	**5.28**	**12.38**	–

Table 16
Results of the Friedman test for the comparisons among the results of the filter UFS methods of Tables 10–15

	Classifier	F_F	Critical value	Significant differences
UCI	SVM	3.509	1.635	Yes
	*k*NN	3.357	1.635	Yes
	NB	2.453	1.635	Yes
ASU	SVM	9.928	1.647	Yes
	*k*NN	7.752	1.647	Yes
	NB	13.641	1.647	Yes

of *k*-means evaluated through the particular evaluation measure. Meanwhile, in Tables 8 and 9, we show the pair-wise comparisons (post hoc comparisons) among the results of the UFS methods over UCI and ASU datasets, respectively. The first column of these tables shows the clustering algorithm along with used evaluation measure; the second column shows the order of the hypothesis,[12] the third column shows the compared methods, the fourth column shows the corresponding *p*-value, and the last one shows an indicator that points out if there is a significant difference or not between the compared filter UFS methods.

From Tables 3 and 4, we can see that in terms of clustering quality the filter UFS methods that get the best results (best average rank) with regard to ACC and NMI were obtained by SVD-Entropy, USFSM, and LS among univariate methods, meanwhile MCFS and MRSF among the multivariate ones. It is noteworthy that some of the aforementioned methods got better results than using the original set of features, as we can see in Fig. 1a and b. Furthermore, based on the first two rows of Table 7, and the post hoc comparisons shown in Table 8, the methods that got the best results are statistically significant compared to the MC filter method, and there is no significant difference among them for both quality measures.

On the other hand, regarding the quality results (ACC and NMI) of the evaluated filter UFS methods on the high dimensional datasets of Table 2. From Tables 5 and 6 we can observe that the best methods were MCFS and NDFS, followed by RRFS and MGSACO among the multivariate methods; meanwhile, LS and SPEC got the best results among the univariate ones. Furthermore, as we can see in the last two rows of Table 7 and the post hoc comparisons shown in Table 9, the results for these best methods are statistically significant compared to the results of JELSR, FSFS, MC, and UDFS for both clustering quality measures.

4.1.2. Results in terms of classification

In a similar way as in the previous subsection, for this evaluation, the results of the assessed filter UFS methods in terms of classification accuracy by using SVM, *k*NN, and NB classifiers, after applying the UFS methods over the datasets of Table 1, are shown in Tables 10–12, while the results for the ASU datasets are in Tables 13–15.

As we can see in Table 10, for datasets of Table 1 the best results in classification performance using SVM were obtained by NDFS followed by MRSF among the multivariate UFS methods; while USFSM and SVD-Entropy got the best results among the univariate ones. This can be verified in Fig. 2a, where it can be seen that NDFS, MRSF, and USFSM are in the first places in all percentages. Moreover, as can be seen in the first row of Table 16 and the post hoc comparisons shown in Table 17, for SVM there are statistically significant differences between the compared filter UFS methods. Specifically, NDFS is significantly better than RRFS, MC, SUD, and FSFS; while MRFS is significantly better than RRFS, MC, and SUD.

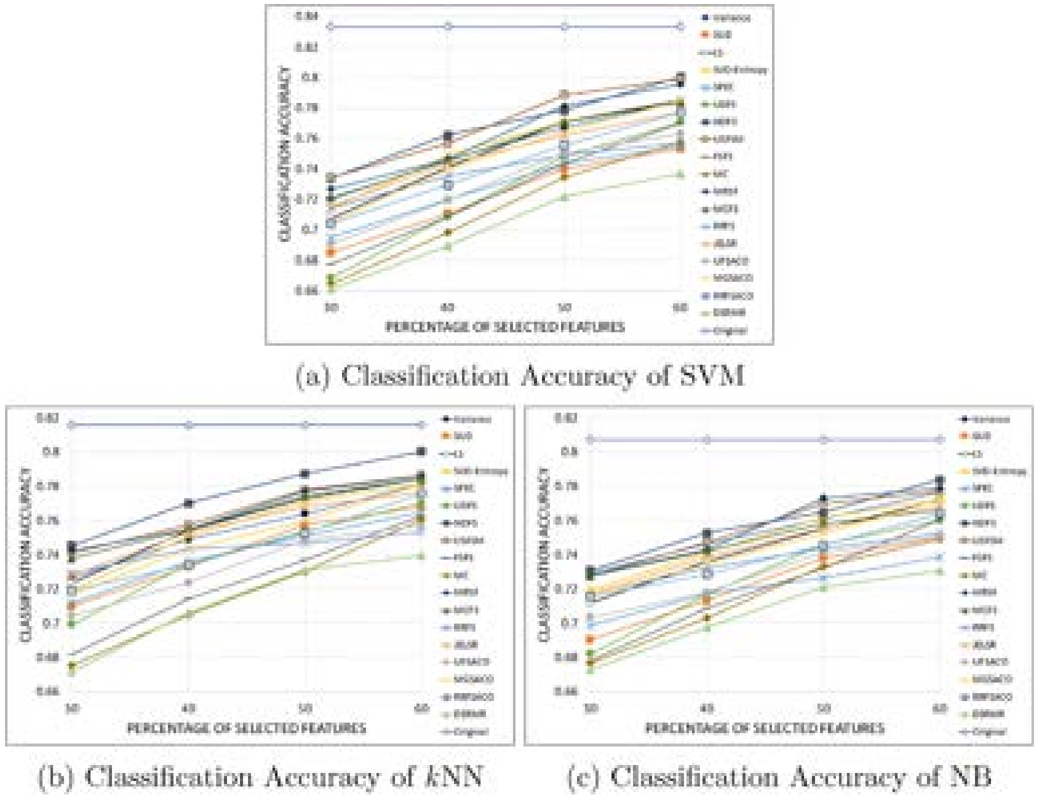
[12] In these tables, we only show the first hypotheses where there is a statistically significant difference.

Table 17
Results (*p*-values) achieved on post hoc comparisons among the results of the filter UFS methods of Tables 10–12 for $\alpha = 0.05$ using SVM, *k*NN, and NB.

i	UFS methods	*p*-value	Significant differences
SVM			
1	NDFS vs. RRFS	0.000005	Yes
2	NDFS vs. MC	0.000018	Yes
3	SUD vs. NDFS	0.000021	Yes
4	MRSF vs. RRFS	0.000098	Yes
5	NDFS vs. FSFS	0.000149	Yes
6	MC vs. MRSF	0.000289	Yes
7	SUD vs. MRSF	0.000334	Yes
8	NDFS vs. DSRMR	0.0004	No
*k*NN			
1	FSFS vs. NDFS	0.000013	Yes
2	NDFS vs. RRFS	0.000016	Yes
3	MC vs. NDFS	0.000019	Yes
4	NDFS vs. UFSACO	0.000024	Yes
5	NDFS vs. DSRMR	0.000568	Yes
6	NDFS vs. RRFSACO	0.000698	Yes
7	SPEC vs. NDFS	0.001874	Yes
8	FSFS vs. MCFS	0.002126	Yes
9	UDFS vs. NDFS	0.002335	Yes
10	USFSM vs. FSFS	0.002485	Yes
11	MCFS vs. RRFS	0.002563	Yes
12	SUD vs. NDFS	0.002643	Yes
13	MC vs. MCFS	0.002898	Yes
14	USFSM vs. RRFS	0.002988	Yes
15	MCFS vs. UFSACO	0.003373	No
NB			
1	MRSF vs. RRFS	0.000224	Yes
2	NDFS vs. RRFS	0.000548	No
3	SPEC vs. MRSF	0.000885	No

Regarding the results of the evaluated filter UFS methods, in terms of classification accuracy by using *k*NN ($k = 3$), in Table 11, we can see that NDFS got again the best results, and it was the best method in all the percentages, as we can see in Fig. 2b. The second best multivariate method was MCFS, meanwhile, USFSM and LS were the best among the univariate ones. Moreover, for this classifier, as it can see in the second row of Table 16 and the post hoc comparisons shown in Table 17, the results of NDFS, MCFS, and USFSM are statistically significant. Specifically, NDFS is significantly better than FSFS, RRFS, MC, UFSACO, DSRMR, RRFSACO, SPEC, UDFS and SUD. For its part, MCFS is significantly better than FSFS, RRFS, MC, and UFSACO. While USFSM is significantly better than FSFS and RRFS.

In the same way, the results of the evaluated filter UFS methods, in terms of classification accuracy by using NB, are shown in Table 12. From this table, we can observe that the best results were obtained by MRSF and NDFS among multivariate methods (as we can corroborate in Fig. 2c); while LS and USFSM got the best results among univariate methods. Moreover, as can be seen in the third row in Table 16 and the post hoc comparisons shown in Table 17, for this classifier, there are also statistically significant differences among the evaluated filter UFS methods, being in this case only MRSF significantly better than RRFS.



(a) Classification Accuracy of SVM

(b) Classification Accuracy of kNN

(c) Classification Accuracy of NB

Fig. 2. Average Classification Accuracy over the results of the compared filter UFS methods using SVM, kNN ($k = 3$), and NB, selecting the 30–60% of features of the UCI datasets of Table 1. The x-axis is the percentage of selected features, and the y-axis is the corresponding average classification accuracy.

Notice that, from Fig. 2, it is possible to observe that the accuracies of all UFS methods increase as the number of selected features increases. When the number of selected features is large enough some of them have similar performance.

On the other hand, the results of the evaluated filter UFS methods, in terms of classification accuracy by using SVM, on the high dimensional datasets of Table 2, are shown in Table 13. From this table, we can see that the bio-inspired methods, MGSACO, UFSACO, and RRFSACO, got the best results followed by RRFS and MCFS among multivariate methods; while USFSM and SPEC got the best results among univariate methods. Moreover, as we can see in the fourth row of Table 16, and the post hoc comparisons of Table 18, the results of the above-mentioned methods are statistically significant.

Similarly, the results of the evaluated filter UFS methods in terms of classification accuracy by using kNN are shown in Table 14. From this table, we can see that the best results on average were achieved by UFSACO, MGSACO, RRFS, MCFS, and RRFSACO, among multivariate methods. Meanwhile, SPEC and USFSM were the best among the univariate ones. Furthermore, from the fifth row in Table 7, and the post hoc comparisons shown in Table 18, we can appreciate that the results for these best methods are statistically significant.

Finally, the results of the evaluated filter UFS methods in terms of classification accuracy by using NB are shown in Table 15. In this table, we can observe that the best results were obtained again by UFSACO, MGSACO, RRFSACO, RRFS, and MCFS among multivariate methods; and USFSM followed by SPEC among the univariate ones. From the last row in Table 16 and the post hoc comparisons shown in Table 18, we can see that the results for these best methods are statistically significant.

4.1.3. Feature selection runtime

In Tables 19 and 20 the runtime (in seconds) spent by each UFS method on each one of the datasets of Tables 1 and 2 is shown, respectively; wherein the rows of Tables 19 and 20 indicate the dataset name, the columns correspond to the feature selection methods, and the last row shows the overall average feature selection runtime. It should be noted that for those methods that could not finish the feature selection in a particular dataset, the symbol "☆" was used.

As we can see in Table 19, regarding the feature selection runtime of filter UFS methods on the datasets of Table 1, the fastest methods were FSFS, Variance, MC, RRFS, MGSACO, UFSACO, and RRFSACO, performing the feature selection in less than one second on average. Meanwhile, SUD, DSRMR, JELSR, and USFSM were the slowest methods on average. On the other hand, regarding the runtime of the evaluated filter UFS methods over the high dimensional datasets of Table 2, from Table 20 we can see that Variance, LS, SPEC, and RRFS were the fastest methods, which performed feature selection in less than 2 seconds on average. Meanwhile, SUD, SVD-Entropy, and DSRMR were the slowest. It should be mentioned that in more than half of the datasets of Table 2 (marked with "☆") SVD-Entropy and SUD could not complete the feature selection after seven days.

4.2. Discussion

As we can see from the results shown in the previous section, in general, no filter-UFS method can be considered the best. However, from our study, it is possible to identify which methods are statistically better in terms of classification and clustering, as well as those with the lowest runtime.

Table 18
Results (p-values) achieved on post hoc comparisons among the results of the filter UFS methods of Tables 13–15 for $\alpha = 0.05$ using SVM, kNN, and NB.

i	UFS methods	p-value	Significant differences
SVM			
1	FSFS vs. MGSACO	0	Yes
2	FSFS vs. UFSACO	0	Yes
3	FSFS vs. RRFSACO	0	Yes
4	FSFS vs. RRFS	0	Yes
5	FSFS vs. MCFS	0	Yes
6	MRSF vs. MGSACO	0	Yes
7	Variance vs. MGSACO	0.000001	Yes
8	USFSM vs. FSFS	0.000001	Yes
9	MRSF vs. UFSACO	0.000001	Yes
10	JELSR vs. MGSACO	0.000002	Yes
11	Variance vs. UFSACO	0.000002	Yes
12	MRSF vs. RRFSACO	0.000006	Yes
13	JELSR vs. UFSACO	0.000008	Yes
14	Variance vs. RRFSACO	0.00001	Yes
16	SVD-Entropy vs. MGSACO	0.00001	Yes
17	MRSF vs. RRFS	0.000015	Yes
18	Variance vs. RRFS	0.000024	Yes
19	MCFS vs. MRSF	0.000025	Yes
20	JELSR vs. RRFSACO	0.000028	Yes
21	SUD vs. MGSACO	0.00003	Yes
22	SVD-Entropy vs. UFSACO	0.00004	Yes
23	Variance vs. MCFS	0.00004	Yes
24	SPEC vs. FSFS	0.000063	Yes
25	RRFS vs. JELSR	0.000067	Yes
26	FSFS vs. NDFS	0.000071	Yes
27	MC vs. MGSACO	0.000099	Yes
28	FSFS vs. UDFS	0.000104	Yes
29	SUD vs. UFSACO	0.00011	Yes
30	MCFS vs. JELSR	0.00011	Yes
31	SVD-Entropy vs. RRFSACO	0.000136	Yes
32	LS vs. MGSACO	0.000152	Yes
33	FSFS vs. DSRMR	0.000178	Yes
34	SVD-Entropy vs. RRFS	0.000299	Yes
35	USFSM vs. MRSF	0.000299	Yes
36	MC vs. UFSACO	0.000331	Yes
37	SUD vs. RRFSACO	0.000349	Yes
38	Variance vs. USFSM	0.000448	No
kNN			
1	Variance vs. UFSACO	0	Yes
2	Variance vs. MGSACO	0	Yes
3	Variance vs. RRFS	0	Yes
4	Variance vs. MCFS	0	Yes
5	JELSR vs. UFSACO	0	Yes
6	JELSR vs. MGSACO	0.000001	Yes
7	Variance vs. RRFSACO	0.000001	Yes
8	RRFS vs. JELSR	0.000002	Yes
9	MCFS vs. JELSR	0.000002	Yes
10	JELSR vs. RRFSACO	0.000006	Yes
11	Variance vs. SPEC	0.000007	Yes
12	FSFS vs. UFSACO	0.000009	Yes
13	FSFS vs. MGSACO	0.000021	Yes
14	SPEC vs. JELSR	0.00003	Yes
15	FSFS vs. RRFS	0.00003	Yes
16	FSFS vs. MCFS	0.000034	Yes
17	Variance vs. USFSM	0.000045	Yes
18	FSFS vs. RRFSACO	0.000099	Yes
19	MRSF vs. UFSACO	0.00016	Yes
20	USFSM vs. JELSR	0.000178	Yes
21	Variance vs. NDFS	0.000331	Yes
22	MRSF vs. MGSACO	0.000349	Yes
23	SPEC vs. FSFS	0.000386	No
NB			
1	FSFS vs. UFSACO	0	Yes
2	FSFS vs. RRFS	0	Yes
3	FSFS vs. RRFSACO	0	Yes
4	FSFS vs. MGSACO	0	Yes
5	Variance vs. UFSACO	0	Yes
6	FSFS vs. MCFS	0	Yes
7	Variance vs. RRFS	0	Yes
8	Variance vs. RRFSACO	0	Yes
9	Variance vs. MGSACO	0	Yes
10	SUD vs. UFSACO	0	Yes

Table 18 (*continued*)

i	UFS methods	*p*-value	Significant differences
11	SUD vs. RRFS	0.000001	Yes
12	SUD vs. RRFSACO	0.000001	Yes
13	SUD vs. MGSACO	0.000001	Yes
14	Variance vs. MCFS	0.000001	Yes
15	JELSR vs. UFSACO	0.000002	Yes
16	UFSACO vs. DSRMR	0.000002	Yes
17	LS vs. UFSACO	0.000002	Yes
18	SVD-Entropy vs. UFSACO	0.000002	Yes
19	RRFS vs. JELSR	0.000002	Yes
20	RRFS vs. DSRMR	0.000002	Yes
21	JELSR vs. RRFSACO	0.000003	Yes
22	RRFSACO vs. DSRMR	0.000003	Yes
23	JELSR vs. MGSACO	0.000003	Yes
24	MGSACO vs. DSRMR	0.000003	Yes
25	LS vs. RRFS	0.000003	Yes
26	SVD-Entropy vs. RRFS	0.000003	Yes
27	LS vs. RRFSACO	0.000004	Yes
28	SVD-Entropy vs. RRFSACO	0.000004	Yes
29	SUD vs. MCFS	0.000004	Yes
30	LS vs. MGSACO	0.000004	Yes
31	SVD-Entropy vs. MGSACO	0.000004	Yes
32	MRSF vs. UFSACO	0.000009	Yes
33	MRSF vs. RRFS	0.000012	Yes
34	MCFS vs. JELSR	0.000013	Yes
35	MCFS vs. DSRMR	0.000013	Yes
36	MRSF vs. RRFSACO1	0.000014	Yes
37	MRSF vs. MGSACO	0.000016	Yes
38	FSFS vs. NDFS	0.000016	Yes
39	LS vs. MCFS	0.000018	Yes
40	SVD-Entropy vs. MCFS	0.000019	Yes
41	USFSM vs. FSFS	0.000027	Yes
42	MCFS vs. MRSF	0.000063	Yes
43	Variance vs. NDFS	0.000099	Yes
44	Variance vs. USFSM	0.00016	Yes
45	SUD vs. NDFS	0.000244	Yes
46	SUD vs. USFSM	0.000386	Yes
47	NDFS vs. JELSR	0.000632	No

For the datasets from the UCI repository that appear in Table 1, which have low and medium dimensionality, we can see that MCFS, MRSF, USFSM, SVD-Entropy, and LS were statistically the best in terms of clustering quality (ACC and NMI), having in some cases even better results than using the whole set of features (original). On the other hand, regarding the supervised classification task, using SVM, *k*NN, and NB, we can see that among the multivariate methods, the best ones were NDFS, MCFS, and MRSF; meanwhile, USFSM was the best among the univariate ones, being those methods statistically better than several of the compared methods, and not having statistically significant differences among them. Moreover, if we take into account the runtime that UFS methods took for performing feature selection, we can say that LS and SVD-Entropy (among the univariate methods) and MCFS and NDFS (among multivariate methods) were the fastest since they performed feature selection in 2.36, 10.03, 4.4, and 13.8 seconds on average, respectively. In summary, the best results in both classification and clustering tasks on small and medium dimensional datasets taken from the UCI repository were obtained by Spectral/Sparse-Learning-based methods. Conversely, multivariate statistical-based methods got the worst results in both tasks; classification and clustering.

On the other hand, for the datasets from the ASU repository that appear in Table 2, which have higher dimensionality than UCI datasets, we can conclude that, in terms of clustering (ACC and NMI), the best methods were NDFS, MCFS, LS, and MGSACO, being those statistically better than several methods and no having statistical differences among them. On the other hand, regarding the results in terms of supervised classification, regardless of the classifier, the best results were obtained by the bio-inspired methods, i.e., UFSACO, MGSACO, and RRFSACO, as well as MCFS, RRFS, and USFSM, being all these methods significantly better than other methods in several cases. However, if we also consider the runtime, RRFS and MCFS methods stand out from the rest, since they had an average runtime of only 1.3 and 9.53 seconds, respectively. Notice that SUD and SVD-Entropy were the slowest methods, which indicate that these methods are not suitable for high dimensional data. In summary, we can say that for high dimensional datasets, the multivariate Spectral/Sparse-Learning-based methods were the best in clustering tasks; meanwhile, for classification tasks, the best results were obtained by the multivariate bio-inspired methods.

Finally, from the results presented in the previous section and the analysis performed, some insights and guidelines for the use of filter UFS methods are given:

- For the low and medium dimensional data, Spectral/Sparse-Learning-based methods are the best option for clustering and classification. Meanwhile, for high dimensional data, bio-inspired methods are a good alternative. However, bio-inspired methods have a higher computational cost than Spectral/Sparse-Learning-based methods. Moreover, in general, Spectral/Sparse-Learning-based methods reach statistically the same results than bio-inspired methods in terms of classification on high dimensional data.

Table 19
Feature selection runtime (in seconds) of filter UFS methods on the UCI datasets of Table 1.

Dataset	*Univariate*						*Multivariate*											
	Variance	SUD	LS	SVD-Entropy	SPEC	USFSM	FSFS	MC	MCFS	MRSF	UDFS	NDFS	RRFS	JELSR	UFSACO	MGSACO	RRFSACO	DSRMR
Inflammations	0.020	0.020	0.013	0.010	0.015	0.124	0.015	0.014	0.025	0.036	0.026	0.129	0.013	0.033	0.014	0.015	0.015	0.722
Audiology	0.044	0.329	0.004	0.099	0.008	1.832	0.033	0.044	0.015	0.066	0.090	0.154	0.043	0.073	0.575	0.573	0.570	4.130
Automovile	0.034	0.125	0.007	0.041	0.007	0.555	0.005	0.024	0.010	0.030	0.070	0.110	0.025	0.142	0.065	0.064	0.065	3.539
Breast-cancer	0.035	0.039	0.004	0.006	0.019	0.544	0.004	0.043	0.019	0.054	0.072	0.129	0.040	0.661	0.057	0.052	0.048	7.317
Contraception	0.160	1.406	0.186	0.026	1.106	87.974	0.005	0.155	0.980	0.737	3.606	2.847	0.170	21.870	0.159	0.168	0.160	2961.286
Credit-approval	0.091	0.609	0.054	0.035	0.247	14.345	0.006	0.086	0.173	0.370	0.522	1.253	0.095	4.684	0.096	0.106	0.093	280.869
Credit-german	0.125	2.080	0.111	0.052	0.577	57.733	0.010	0.117	0.403	0.420	2.507	1.928	0.124	9.065	0.138	0.147	0.137	834.277
Cylinder-bands	0.223	1.733	0.035	0.204	0.287	18.988	0.029	0.210	0.192	0.439	0.810	1.105	0.221	2.299	0.363	0.374	0.350	160.887
Dermatology	0.083	0.595	0.025	0.165	0.074	5.323	0.021	0.081	0.080	0.169	0.392	0.615	0.079	1.331	0.156	0.162	0.158	31.873
Ecoli	0.044	0.245	0.019	0.023	0.070	0.002	0.050	0.056	0.045	0.053	0.507	0.845	0.859	0.046	0.045	0.048	0.001	20.771
Flags	0.093	0.133	0.027	0.067	0.045	0.812	0.014	0.091	0.035	0.081	0.247	0.524	0.090	0.391	0.148	0.153	0.143	4.090
Heart-c	0.046	0.105	0.012	0.025	0.045	1.143	0.004	0.047	0.054	0.061	0.120	0.449	0.047	0.853	0.051	0.049	0.051	15.620
Heart-h	0.043	0.097	0.026	0.025	0.060	1.035	0.004	0.043	0.051	0.044	0.126	0.449	0.054	0.703	0.052	0.053	0.051	14.918
Heart-statlog	0.039	0.436	0.023	0.018	0.049	0.891	0.004	0.040	0.043	0.068	0.091	0.430	0.043	0.660	0.053	0.048	0.037	11.016
Hepatitis	0.043	0.053	0.017	0.057	0.008	0.293	0.005	0.032	0.028	0.062	0.154	0.376	0.035	0.261	0.055	0.054	0.058	2.863
Horse-colic	0.056	0.335	0.014	0.056	0.077	3.398	0.010	0.061	0.065	0.122	0.337	0.600	0.067	1.065	0.088	0.082	0.091	33.930
Hypothyroid	0.411	61.574	1.717	0.123	9.668	158.054	0.034	0.450	3.331	4.228	24.067	12.641	0.420	361.974	0.504	0.502	0.503	☆
Ionosphere	0.053	5.559	0.090	0.260	0.206	6.032	0.017	0.059	0.158	0.338	0.663	0.906	0.069	2.233	0.151	0.149	0.152	16.507
Iris	0.032	0.018	0.085	0.075	0.070	0.074	0.002	0.035	0.098	0.033	0.040	0.562	0.036	0.465	0.026	0.028	0.032	2.941
Liver-disorders	0.049	0.186	0.065	0.167	0.158	0.854	0.002	0.052	0.111	0.150	0.114	0.893	0.051	1.468	0.041	0.050	0.052	10.153
Lung-cancer	0.067	0.017	0.003	0.253	0.035	0.038	0.039	0.058	0.050	0.223	0.218	0.608	0.078	0.115	0.491	0.492	0.496	0.080
Lymphography	0.041	0.047	0.108	0.160	0.093	0.279	0.007	0.035	0.082	0.140	0.231	0.668	0.042	0.452	0.047	0.049	0.055	1.389
Promoters	0.096	0.130	0.013	0.566	0.060	0.485	0.045	0.106	0.219	0.443	0.382	0.661	0.094	0.258	0.519	0.542	0.552	0.726
Monks-1-train	0.025	0.006	0.010	0.132	0.087	0.056	0.002	0.029	0.059	0.060	0.038	0.531	0.024	0.258	0.036	0.034	0.029	0.534
Monks-2-train	0.038	0.012	0.063	0.065	0.085	0.115	0.002	0.041	0.083	0.088	0.060	0.518	0.037	0.529	0.033	0.036	0.030	2.253
Monks-3-train	0.039	0.008	0.031	0.068	0.074	0.046	0.002	0.027	0.064	0.083	0.051	0.500	0.026	0.348	0.027	0.026	0.032	0.806
Parkinsons	0.051	0.714	0.080	0.238	0.124	0.668	0.008	0.057	0.082	0.274	0.223	0.527	0.039	0.628	0.067	0.058	0.053	3.377
Pima	0.089	1.533	0.221	0.106	0.624	11.124	0.005	0.094	0.378	0.489	0.594	2.086	0.095	5.293	0.094	0.089	0.093	311.851
Post-Operative	0.035	0.005	0.002	0.051	0.063	0.044	0.007	0.024	0.075	0.064	0.037	0.354	0.026	0.177	0.027	0.031	0.024	0.416
Segment	0.241	85.148	0.748	0.214	5.529	31.369	0.013	0.263	1.734	50.697	19.391	9.650	0.240	173.343	0.279	0.280	0.261	☆
Sonar	0.064	6.098	0.121	0.911	0.143	3.727	0.062	0.058	0.188	0.461	0.462	0.851	0.051	0.668	0.557	0.580	0.579	5.000
Soybean	0.061	1.289	0.031	0.044	0.084	15.185	0.024	0.063	0.070	0.111	0.242	0.388	0.053	0.744	0.125	0.121	0.125	207.900
Sponge	0.056	0.033	0.003	0.046	0.005	0.108	0.064	0.087	0.099	0.346	0.093	0.273	0.089	0.401	0.285	0.316	0.281	0.778
Tae	0.031	0.018	0.091	0.085	0.126	0.091	0.009	0.032	0.076	0.087	0.054	0.541	0.028	0.346	0.024	0.031	0.027	2.211
Thoracic	0.081	0.314	0.252	0.161	0.469	5.061	0.007	0.066	0.193	0.378	0.412	1.432	0.080	2.945	0.077	0.069	0.078	53.425
Tic-tac-toe	0.103	0.613	0.223	0.148	0.712	26.206	0.012	0.123	0.691	0.466	1.252	2.639	0.113	9.307	0.122	0.141	0.112	587.018
Silhouettes	0.103	9.078	0.406	0.156	0.471	34.079	0.007	0.101	0.453	0.630	2.243	2.554	0.095	10.213	0.109	0.112	0.117	422.910
Vote	0.057	0.270	0.151	0.230	0.259	4.257	0.004	0.064	0.231	0.412	0.465	0.950	0.061	2.719	0.079	0.100	0.078	29.158
Vowel	0.116	1.096	0.197	0.264	0.748	39.833	0.005	0.133	1.098	1.290	1.565	3.438	0.135	20.952	0.118	0.124	0.135	728.950
Wdbc	0.071	11.486	0.193	0.297	0.475	18.530	0.021	0.076	0.409	0.666	1.085	1.845	0.069	5.805	0.145	0.127	0.155	114.834
Wine	0.043	0.207	0.113	0.066	0.104	0.315	0.003	0.041	0.163	0.078	0.173	0.579	0.041	0.458	0.044	0.035	0.039	2.602
Zoo	0.045	0.025	0.009	0.159	0.097	0.109	0.004	0.048	0.068	0.127	0.277	0.514	0.055	0.204	0.052	0.062	0.064	0.869
Splice	0.620	146.204	0.453	0.323	3.701	282.399	0.138	0.629	0.327	0.631	3.830	4.930	0.600	262.076	0.990	1.002	0.999	54141.773
Spambase	0.478	3093.478	6.462	10.988	49.921	869.086	0.213	0.618	7.489	13.385	73.327	27.157	0.627	676.102	1.115	1.127	1.092	☆
Mushroom	0.911	319.597	10.304	4.494	189.602	1669.739	0.067	0.987	14.918	18.676	172.111	85.428	0.995	12952.449	0.921	0.903	0.941	☆
Optdigits	0.589	6551.997	11.059	16.062	112.459	1840.130	0.506	0.849	6.319	10.339	108.962	52.133	0.663	556.109	1.480	1.447	1.574	☆
Waveform-5000	0.654	2200.920	6.226	4.216	59.868	805.023	0.153	0.610	5.585	7.705	78.531	32.036	0.680	424.334	0.828	0.823	0.906	☆
Pendigits	1.412	1954.795	19.685	7.350	400.148	3097.380	0.115	1.366	86.867	85.778	244.185	172.239	1.345	12455.331	1.298	1.344	1.307	☆
Clean2	0.948	88162.674	35.251	451.851	265.930	14429.960	2.638	2.249	49.570	5695.949	199.024	73.753	1.037	5656.904	10.069	10.366	9.136	☆
Nursery	1.419	☆	23.097	0.652	933.319	☆	0.061	1.532	40.408	65.965	295.421	184.206	1.403	☆	1.333	1.484	1.452	☆
Average	**0.206**	**2094.357**	**2.363**	**10.038**	**40.764**	**480.519**	**0.090**	**0.244**	**4.479**	**119.272**	**24.790**	**13.819**	**0.230**	**686.321**	**0.485**	**0.496**	**0.472**	**1488.697**

Table 20
Feature selection runtime (in seconds) of filter UFS methods on the ASU datasets of Table 2.

Dataset	*Univariate*						*Multivariate*											
	Variance	SUD	LS	SVD-Entropy	SPEC	USFSM	FSFS	MC	MCFS	MRSF	UDFS	NDFS	RRFS	JELSR	UFSACO	MGSACO	RRFSACO	DSRMR
Yale	0.209	1521.626	0.113	21.094	0.060	23.568	11.148	0.333	2.000	4.537	4.287	2.773	0.185	1.046	182.861	191.384	171.725	31.371
Isolet	0.210	45274.904	0.910	421.204	2.843	895.093	4.687	2.223	6.194	443.741	5.925	3.355	0.301	20.296	79.237	96.437	88.899	10684.951
ORL	0.197	8942.702	0.065	134.566	0.111	168.724	11.200	0.577	3.018	31.862	4.407	3.047	0.199	4.937	176.422	211.914	187.965	344.950
WarpPIE10P	0.411	15148.739	0.071	476.814	0.107	170.650	29.941	1.471	2.543	65.394	45.678	17.016	0.433	14.794	599.075	596.172	549.380	191.877
Colon	0.330	1138.859	0.019	821.076	0.011	11.366	26.154	0.563	1.888	0.516	23.522	9.608	0.331	8.257	436.344	577.357	596.411	34.855
WarpAR10P	0.632	6927.235	0.087	329.795	0.062	66.802	52.959	4.745	5.244	1.893	46.065	28.691	0.544	11.872	822.182	692.148	629.987	98.522
Lung	0.715	29232.301	0.202	564.480	0.145	294.234	115.551	7.663	5.465	1.766	128.470	61.567	0.746	18.441	931.764	1037.636	885.322	217.789
COIL20	0.297	74535.114	1.152	389.663	3.853	1645.114	15.869	7.745	8.773	17.710	9.900	5.991	0.319	9.924	193.336	249.439	234.778	6598.594
Lymphoma	1.011	5540.365	0.084	8188.257	0.032	96.359	123.516	5.137	4.215	1.414	235.397	108.863	0.909	32.609	1184.912	1120.963	1132.613	213.736
GLIOMA	1.216	2010.759	0.153	6745.593	0.016	35.962	172.575	3.911	5.150	1.286	315.086	152.249	1.079	37.534	1395.694	1641.870	1383.129	211.850
ALLAML	1.923	45653.793	0.748	43259.879	0.038	166.702	557.268	11.467	8.949	1.835	1262.915	597.034	2.132	207.318	2483.426	3042.101	2844.765	1090.328
CLL_SUB_111	4.342	☆	0.581	☆	0.106	962.483	2650.784	60.624	17.745	5.024	3726.788	3540.647	2.821	685.348	4345.196	5222.183	3454.412	3962.631
Carcinom	2.267	☆	0.584	☆	0.368	1458.842	1140.651	57.548	16.682	8.918	2102.941	934.011	2.497	409.283	3783.277	4036.756	4276.910	2452.996
GLI_85	4.204	☆	0.649	☆	0.141	2157.922	7356.696	94.912	12.227	6.703	32566.062	20313.540	3.750	4227.078	8283.713	8773.398	0.006	28303.351
Prostate_GE	1.053	☆	0.095	☆	0.098	299.956	305.371	12.906	5.413	5.590	799.925	201.379	1.393	140.985	1857.901	1981.829	1502.772	585.752
SMK_CAN_187	3.372	☆	0.528	☆	0.976	8669.982	4947.196	131.492	22.888	14.955	30216.412	8484.230	3.412	3643.437	6741.467	8250.373	7716.223	15078.189
TOX_171	0.967	☆	0.151	☆	0.183	583.957	214.705	10.803	4.118	3.926	677.928	177.107	0.972	162.001	1464.461	3180.115	2525.845	624.122
Arcene	2.374	☆	0.522	☆	0.372	2336.654	785.669	48.041	28.035	54.659	2599.092	1322.105	1.689	561.996	4709.873	5204.064	4091.641	2434.245
Leukemia	1.217	☆	0.150	☆	0.040	168.021	303.200	4.779	4.287	1.811	1222.528	312.825	1.172	85.149	2324.686	2903.682	2064.330	578.998
Nci9	1.601	☆	0.092	☆	0.045	226.447	653.958	7.097	5.467	3.960	2429.669	762.553	1.606	198.345	3240.901	3352.388	3111.060	1337.468
Orlraws10P	1.710	☆	0.142	☆	0.082	632.948	739.507	11.660	6.435	4.899	2811.730	915.016	1.712	226.522	3995.451	4277.007	3361.445	1613.027
Pixraw10P	1.664	☆	0.138	☆	0.120	594.508	683.231	11.061	7.180	2.963	2616.453	823.172	1.671	207.988	2950.954	4283.712	3198.680	1504.253
RELATHE	1.115	☆	3.735	☆	15.291	18124.805	218.749	250.778	17.589	41.658	316.316	103.702	1.136	46.592	1685.321	1882.767	1468.031	9519.645
PCMAC	1.021	☆	9.581	☆	21.304	9169.304	150.886	199.030	16.108	44.702	162.280	60.179	0.995	35.825	1213.921	1339.683	1165.970	16477.637
BASEHOCK	1.484	☆	14.158	☆	☆	590.560	329.797	520.389	20.832	☆	☆	125.680	1.492	59.700	2161.037	2430.799	2067.499	19921.239
Average	**1.422**	**21447.854**	**1.388**	**5577.493**	**1.934**	**2040.017**	**864.051**	**58.678**	**9.538**	**32.155**	**3513.741**	**1562.654**	**1.340**	**442.291**	**2289.736**	**2663.047**	**1948.392**	**4964.495**

- Filter Statistical-based methods (Variance, FSFS, MC, and RRFS) were the fastest in low and medium dimensional data. However, these methods usually provide the worst results in terms of clustering and classification quality.
- In general, in all the used datasets, multivariate Spectral/Sparse-Learning-based methods (MCFS, NDFS, and MRFS) and univariate Spectral-based methods (USFSM, SPEC, and LS) are significantly better than statistical-based methods (MC and FSFS) in terms of classification and clustering quality. Furthermore, if we take into account the runtime, MCFS was the best due to its good compromise between quality and efficiency.

5. Concluding remarks and future work

Unsupervised Feature Selection methods have drawn interest in several research areas due to their practical significance and capability to select features in unlabeled data (unsupervised datasets). Specifically, UFS methods belonging to the filter approach have gained considerable interest due to their efficiency, scalability, and quality of selection in expert and intelligent systems applications.

In this paper, we performed an empirical and systematic evaluation of 18 of the most relevant and recent filter UFS methods of the state-of-the-art. The performance of the UFS methods was evaluated in terms of clustering, classification, and runtime on 50 low and medium dimensional datasets taken from the UCI Machine Learning Repository as well as 25 high dimensional datasets taken from the ASU Feature Selection Repository. The quality of the results in the clustering tasks was measured in terms of ACC and NMI using *k*-means. Meanwhile, the quality of the results for supervised classification evaluations was measured in terms of Classification Accuracy using SVM, *k*NN, and NB. Furthermore, the significance of the results was confirmed by applying the Friedman test and Holm post hoc procedure. Finally, we provided a general discussion based on the results of the compared filter UFS methods, and we suggested some practical guidelines and insights for their use (see Section 4.2).

An interesting finding in our study is that Unsupervised Feature Selection helps to improve the quality results in terms of both classification and clustering. However, it is possible to observe that the quality of the results in terms of clustering is better than the quality of the results in terms of supervised classification. This was expected since, in clustering, UFS methods usually are able to identify those features that define cluster structures; however, these clusters structures do not necessarily correspond to the classification defined in the supervised datasets, so the results in terms of classification quality tend to be worse. Another interesting finding is that we confirm that there is no filter UFS method that can be considered the best. However, in general, for the low and medium and high dimensional data, Spectral/Sparse-Learning-based methods arise as the best option for clustering and classification.

Of course, like in any empirical study, there are some limitations in our experiments. On the one hand, feature selection evaluation was made only on supervised datasets using external evaluation measures for assessing the clustering quality. However, even though this is a widely used evaluation strategy, to the best of our knowledge, to define the best way to evaluate UFS methods is still an open problem. On the other hand, we are aware that our study is empirical. Nevertheless, we are sure that it may serve as a basis for theorizing, and defining future theoretical research lines into the Unsupervised Feature Selection.

Finally, future work of this research includes the following:

- In our experiments, for evaluating filter UFS methods, since the optimal number of features to be selected for each method for a specific dataset is difficult to know, the number of features to select was chosen arbitrarily. In this regard, to define a strategy for automatic selection of the best number of features for each method could be studied.
- As we can realize, the amount of literature in Unsupervised Feature Selection is very extensive, and continuously new works are introduced. Hence, a similar study including UFS based on wrapper and hybrid approaches also might be interesting.
- Currently, there are no software tools that allow performing a comprehensive comparative analysis of UFS methods. Therefore, the development of a public and extensible software platform to facilitate the assessment, analysis and statistical comparisons of UFS methods would be really useful for the research community.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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